

High-Dimensional Bayesian Forecast Combination

Candidate Number: RRBL3

Supervisor: Prof. Jim Griffin

Faculty of Mathematical & Physical Sciences

Department of Statistical Science

University College London

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Abstract

Forecast combination is valuable because the performance of individual forecasters varies over time, and combining forecasts is usually more accurate than selecting a single best forecaster. Unlike a point forecast, a predictive distribution quantifies uncertainty around a forecast, which a decision-maker can use to assess risk. This thesis studies how a decision-maker should combine point forecasts from many individual forecasters into a single predictive distribution. A central paper in this area, [Winkler \[1981\]](#), proposes a Bayesian solution that requires the precision matrix (the inverse of the covariance matrix) of the forecasters' joint error distribution. Given an estimated precision matrix, forecasters that are more precise and less correlated with others typically receive a larger weight. However, it becomes difficult to apply this technique once the number of forecasters is large relative to the amount of data available, because the precision matrix has too many free parameters to estimate reliably. We address this by placing a Bayesian graphical lasso prior [\[Wang, 2012\]](#) on the precision matrix, which shrinks many off-diagonal entries toward zero and reduces the effective dimension of the estimation problem. This approach reduces the mean squared forecast error (MSFE) to about 14% of the sample covariance benchmark in an application to the ECB Survey of Professional Forecasters ($p = 35$). We also test an extension that extracts common factors from the forecast errors before applying the graphical lasso, but we do not find evidence of an underlying factor structure. Two further extensions, correcting for forecaster bias and allowing the precision matrix to vary over time, each provide a modest improvement in forecast accuracy.

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1 | Introduction and Background

A classic experiment related to my subject is conducted by showing a large number of participants a glass jar containing a substantial amount of pennies. Each individual is asked to estimate the number of pennies in the jar. When analyzing the given predictions, one will observe that individuals perform very poorly at this task, but the average prediction of the crowd is surprisingly accurate. The participants usually share a systematic bias, which can be interpreted as a shared misconception about the true value. But, each participant also tends to make some ‘personal’ error that is independent of the errors of others. By averaging the individual predictions, these independent errors cancel out asymptotically, leading to a more accurate estimate of the number of pennies. This simple yet powerful idea of combining predictions to obtain a more accurate measure dates back to at least [Galton \[1907\]](#).

Viewed from a statistical perspective, real-world problems require us to make statements about *unknown* values of a random variable that are currently *uncertain*. The real data generating process (DGP) of a random variable is often complex; it can be affected by time-varying trends, seasonality, heteroskedasticity and structural breaks [[Clements and Hendry, 1998](#)]. Diversifying forecast errors allows us to reduce the uncertainty in our views on the underlying DGP, although any systematic bias shared between forecasters will persist regardless [[Bates and Granger, 1969](#)].

When combining multiple forecasts of the same unknown value, a researcher is broadly faced with three questions: (i) how reliable are the individual forecasts at hand, (ii) how should they be weighted relative to one another, and (iii) how should uncertainty in the chosen combination be quantified? Surprisingly, past literature has often neglected (iii), despite point forecasts alone being insufficient for decision-making [[Wang et al., 2023](#)].

A natural framework for addressing all three concerns simultaneously is that of Bayesian statistics. Instead of mechanically aggregating individual forecasts, the Bayesian approach interprets each individual forecast as carrying information about the unknown value, and updates a prior belief in light of this information to form a posterior. The quality of individual forecasts is reflected in how much the posterior shifts toward them; the combination weights arise natu-

rally from the posterior rather than being imposed arbitrarily; and uncertainty in the combined forecast is represented explicitly through the posterior distribution rather than collapsed into a point estimate [Bunn, 1975].

Translating this Bayesian idea into a practical procedure is, however, not straightforward. Existing combination frameworks either require full predictive densities as input from each forecaster, or return only a single point forecast as output. Both approaches can be limiting in practice. First, individual forecasters often report only a point forecast, which rules out combination schemes that require full predictive densities as input. Second, economic agents often need uncertainty quantification around a point forecast: a policymaker reading GDP growth projections is less interested in the combination itself than in the likelihood of a recession or the probability of extreme outcomes. Meanwhile, forecasting panels have recently grown considerably as data availability has increased. The ECB Survey of Professional Forecasters, for instance, now regularly reports forecasts from over fifty individual respondents each quarter. A practical framework should therefore be able to take point forecasts as inputs, return a full predictive distribution as output, and scale reliably and efficiently to large panels.

To formally introduce some notation, consider a panel of p forecasters observed over a time series of T periods, each reporting a point forecast $\mu_{i,t}$ of some unknown quantity θ_t in period t , for $i = 1, \dots, p$ and $t = 1, \dots, T$. The forecast error of forecaster i in period t is $u_{i,t} = \mu_{i,t} - \theta_t$. Assuming forecast errors are independently and identically distributed over time, we define Σ as the covariance matrix of the forecast errors $u_{1,t}, \dots, u_{p,t}$: its diagonal entries represent the variances of each forecaster's error, and its off-diagonal entries represent the covariances between each pair of forecasters' errors. The precision matrix of forecast errors is defined as $\Omega = \Sigma^{-1}$. Intuitively, the precision matrix Ω captures how each pair of forecast errors relates, once the influence of all other forecasters in the panel is removed. A zero entry ω_{ij} of Ω implies that forecasters i and j 's errors are conditionally independent, that is, independent once we know the errors of all other forecasters. A precision matrix with many such zero entries is called 'sparse'.

[Winkler \[1981\]](#) stands out as a natural Bayesian reference point among existing approaches: given only individual point forecasts as inputs, it provides a closed-form procedure for combining

them into a complete predictive distribution, satisfying precisely the criteria outlined above. The procedure derives both the combination weights and the predictive uncertainty from a single object: the precision matrix of forecast errors $\mathbf{\Omega}$. This implies that questions (i), (ii) and (iii) can be resolved simultaneously within a single coherent model, an unusual property in the forecast combination literature [Wang et al., 2023].

The framework, however, relies on strong assumptions: forecasters are assumed to be unbiased, forecast errors are assumed to be stationary, and the precision matrix of forecast errors is assumed to be known, or at least estimable, despite the number of free parameters growing quadratically with the size of the forecast panel. The goal of this thesis is to examine and relax some of these underlying assumptions, evaluate the consequences theoretically and empirically, and in doing so develop a practical Bayesian framework for forecast combination that is useful to applied forecasters with only point forecasts at hand. Relaxing assumptions is not only a technical exercise, it also requires a careful examination of the dependence structure of forecast errors. Are forecasters conditionally independent? Is the precision matrix of forecast errors sparse once we account for common shocks affecting the entire panel? Does the error structure evolve over time? Are professional forecasters systematically biased? This thesis aims to tackle such questions through simulation studies and an empirical macro-economic application.

Related Literature

The literature on forecast combination has developed along two main strands, distinguished by whether the objects being combined are point forecasts or full predictive distributions. The first strand, dating back to Bates and Granger [1969], treats combination as an averaging problem: individual point forecasts are combined into a single point forecast, with weights derived from the forecasters' historical performance. This strand has produced a number of weighting schemes, ranging from the simple equal-weighted average to Granger and Ramanathan [1984]'s regression-based combination weights. Notably, the equal-weighted average frequently outperforms more sophisticated weighting schemes in practice, a finding referred to as the forecast combination puzzle [Smith and Wallis, 2009]. The second strand combines full predictive distributions directly, treating each forecaster's density as the input to be combined, as in Bayesian

model averaging [Hoeting et al., 1999] and density combination methods such as McAlinn and West [2019]. Wang et al. [2023] provide a comprehensive review of both strands. Winkler [1981]’s framework, introduced in Section 1.1, differs from both: it takes point forecasts as input, as in the first strand, but produces a full predictive distribution as output, as in the second. This combination has received relatively little attention in the literature. Jouini and Clemen [1996] offer a comparable approach using copulas.

A separate body of work is relevant once we ask how the precision matrix $\mathbf{\Omega}$ itself should be estimated in practice, since it is the single object that Winkler [1981]’s framework depends on. When the number of forecasters p grows, the sample covariance matrix becomes an increasingly unreliable estimate of $\mathbf{\Sigma}$, since the number of free parameters grows quadratically in p whilst T is fixed by how much historical data is available [Ledoit and Wolf, 2004]. A natural way to simplify the estimation problem would be to impose sparsity on $\mathbf{\Sigma}$, or directly on $\mathbf{\Omega}$. Fixing many entries to zero reduces the number of free parameters that need to be estimated. However, imposing sparsity on $\mathbf{\Sigma}$ would mean that most pairs of forecasters’ errors are entirely unrelated, which is empirically unrealistic: forecasters often share common beliefs and are exposed to similar information, which means that most error pairs are at least somewhat correlated. Sparsity on $\mathbf{\Omega}$ instead imposes conditional independence: most pairs of forecasters’ errors are unrelated once we account for everyone else in the panel. Empirically, precision matrices tend to be sparser than covariance matrices: conditional independence between two forecasters, once the rest of the panel is accounted for, is more common than marginal independence. Friedman et al. [2008] formalized this idea in the frequentist graphical lasso, which penalizes the likelihood of forecast errors to shrink many entries of $\mathbf{\Omega}$ to exactly zero. Wang [2012] developed a Bayesian counterpart, which places a Laplace prior on the off-diagonal entries of $\mathbf{\Omega}$ to impose shrinkage.

The work that sits closest to this thesis is that of Lee and Seregina [2025], who use the graphical lasso to estimate the precision matrix of forecast errors $\mathbf{\Omega}$ in a large panel, and plug the resulting estimate into the same weights formula as Winkler [1981] and Bates and Granger [1969]. Their approach, however, is entirely frequentist: the graphical lasso returns a single point estimate of $\mathbf{\Omega}$, so the resulting combination weights carry no associated measure of uncertainty. This thesis instead places a Bayesian graphical lasso prior on the precision matrix within Winkler’s

framework, so that estimation uncertainty in $\mathbf{\Omega}$ is carried through to the forecast combination.

The remainder of this thesis is organized as follows. Chapter 2 develops the methodological extensions to Winkler’s framework in detail: the graphical lasso prior, the Bayesian factor graphical lasso, bias correction and non-stationarity. Chapter 3 evaluates these extensions in a controlled simulation study, where the true data-generating process is known and estimators can be compared directly. Chapter 4 applies the framework to the ECB Survey of Professional Forecasters, combining GDP growth forecasts from a panel of professional forecasters, finding that a sparse Bayesian approach substantially reduces forecast error and improves uncertainty quantification relative to a simple baseline. Chapter 5 concludes, addresses limitations, and discusses directions for future work.

The remainder of this chapter prepares the foundation for these extensions. Section 1.1 introduces Winkler’s framework: the setup, the closed-form result, and the assumptions it relies on. Section 1.2 then turns to the practical question of how the precision matrix on which Winkler’s framework relies should actually be obtained. Two routes are available, and we explain why this thesis uses one over the other. Section 1.3 closes by going through the modeling assumptions of Winkler’s framework in turn, and presenting how some will be relaxed in Chapter 2.

1.1 The Winkler Framework

In the framework that will be introduced, forecast errors are assumed to be identically distributed across periods, so that a single period is representative of every other. For the remainder of this section we therefore drop the time index t , purely for notational convenience: the reader should keep in mind that what follows still concerns a time series of forecasts, and that the same derivation applies in every period.

Suppose that p forecasters each report a point forecast μ_i for some unknown quantity θ . Each forecast produces an error $u_i = \mu_i - \theta$, and these errors are collected into a vector $\mathbf{u} = (u_1, \dots, u_p)'$. The errors are modeled as multivariate normal with mean zero and covariance matrix $\mathbf{\Sigma}$,

$$\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}), \tag{1.1}$$

where the zero-mean assumption reflects that forecasters are assumed to be unbiased, and Σ captures both the precision of each individual forecaster and the dependence between their forecast errors. Given Σ , the consensus forecast is a weighted average that can simply be denoted as $\hat{\theta} = \mathbf{w}'\boldsymbol{\mu}$. The optimal weights are given by the following result.

Proposition 1 (Winkler, 1981). *Under assumption (1.1) and a diffuse prior $p(\theta) \propto 1$, the posterior for θ given $\boldsymbol{\mu}$ is Gaussian with mean and variance*

$$\tilde{\mu} = \frac{\mathbf{e}'\boldsymbol{\Omega}\boldsymbol{\mu}}{\mathbf{e}'\boldsymbol{\Omega}\mathbf{e}}, \quad \tilde{\sigma}^2 = \frac{1}{\mathbf{e}'\boldsymbol{\Omega}\mathbf{e}}, \quad (1.2)$$

where $\boldsymbol{\Omega} = \Sigma^{-1}$ and $\mathbf{e} = (1, \dots, 1)'$. The posterior mean is a weighted average of the individual forecasts with weights $\mathbf{w} = \mathbf{e}'\boldsymbol{\Omega}/\mathbf{e}'\boldsymbol{\Omega}\mathbf{e}$, and $\tilde{\sigma}^2$ quantifies the remaining uncertainty after combination.

A brief remark for readers that are not familiar with Bayesian statistics: before seeing any forecasts, we start with a *prior* belief about θ . Here we use a flat prior, $p(\theta) \propto 1$, which implies that we assume nothing about θ before seeing the data. After observing the panel's forecasts $\boldsymbol{\mu}$, this belief is updated into a *posterior* belief, which is the $\mathcal{N}(\tilde{\mu}, \tilde{\sigma}^2)$ distribution above. The centre of this posterior distribution, $\tilde{\mu}$, is exactly our final combined forecast $\hat{\theta}$, the weighted average $\mathbf{w}'\boldsymbol{\mu}$ given above. Combining forecasts, in other words, is nothing more than locating the centre of this posterior. The posterior distribution's variance $\tilde{\sigma}^2$ quantifies the uncertainty around that centre.

Proof. We decompose the reported point forecasts by specifying their deviation from the true value $\boldsymbol{\mu} = \theta \mathbf{e} + \mathbf{u}$. Now, the likelihood of observing $\boldsymbol{\mu}$ given θ is

$$p(\boldsymbol{\mu} \mid \theta) \propto \exp\left\{-\frac{1}{2}(\boldsymbol{\mu} - \theta\mathbf{e})' \boldsymbol{\Omega} (\boldsymbol{\mu} - \theta\mathbf{e})\right\}.$$

Under a diffuse prior $p(\theta) \propto 1$, the posterior $p(\theta \mid \boldsymbol{\mu})$ is proportional to the likelihood. Expanding

the exponent,

$$\begin{aligned}(\boldsymbol{\mu} - \theta \mathbf{e})' \boldsymbol{\Omega} (\boldsymbol{\mu} - \theta \mathbf{e}) &= \boldsymbol{\mu}' \boldsymbol{\Omega} \boldsymbol{\mu} - 2\theta \mathbf{e}' \boldsymbol{\Omega} \boldsymbol{\mu} + \theta^2 \mathbf{e}' \boldsymbol{\Omega} \mathbf{e} \\ &= (\mathbf{e}' \boldsymbol{\Omega} \mathbf{e}) \left(\theta - \frac{\mathbf{e}' \boldsymbol{\Omega} \boldsymbol{\mu}}{\mathbf{e}' \boldsymbol{\Omega} \mathbf{e}} \right)^2 + C,\end{aligned}$$

where C collects terms that do not depend on θ . The posterior is therefore

$$p(\theta \mid \boldsymbol{\mu}) \propto \exp \left\{ -\frac{(\theta - \tilde{\mu})^2}{2\tilde{\sigma}^2} \right\},$$

which is the kernel of a $\mathcal{N}(\tilde{\mu}, \tilde{\sigma}^2)$ density with $\tilde{\mu}$ and $\tilde{\sigma}^2$ as in (1.2). Writing $\tilde{\mu} = \mathbf{w}' \boldsymbol{\mu}$ with weights $w_i = \sum_j \Omega_{ij} / \mathbf{e}' \boldsymbol{\Omega} \mathbf{e}$, the weights sum to one since the numerator $\sum_i w_i = \sum_i \sum_j \Omega_{ij} = \mathbf{e}' \boldsymbol{\Omega} \mathbf{e}$. $\square$

Proposition 1 delivers a surprisingly simple answer to questions (i), (ii), and (iii) from the introduction: all three are resolved by a single object, the precision matrix $\boldsymbol{\Omega}$. The weights $\mathbf{w}$ are determined entirely by $\boldsymbol{\Omega}$, reflecting both individual forecaster reliability and the dependence between their errors. The posterior variance $\tilde{\sigma}^2$ also follows directly from the precision matrix.

The result, however, relies on a number of assumptions. Equation (1.1) already states two of these assumptions directly: that forecast errors have mean zero (A1) and are normally distributed (A2). Proposition 1 imposes two more implicitly: it treats forecast errors as identically distributed over time (A3), and $\boldsymbol{\Sigma}$ as known (A4). Finally, Proposition 1 places no restriction whatsoever on the structure of $\boldsymbol{\Omega}$ itself, which is a gap that will be further studied in Chapter 2.

A1. Unbiasedness.

$\mathbb{E}[\mathbf{u}] = \mathbf{0}$, as already assumed in (1.1). It is documented that some professional forecasting panels violate this assumption [e.g., Genre et al., 2013].

A2. Normality.

$\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, as already assumed in (1.1), ignoring the possibility of skewed or heavy-tailed errors.

A3. Independently and identically distributed errors across time.

A1 and A2 above are stated for a single period. Dropping the time index t throughout this

section relies on forecast errors being identically distributed across periods, so that Σ is stationary over time. Throughout this thesis we assume the stronger property that forecast errors are independently and identically distributed, $\mathbf{u}_t \stackrel{iid}{\sim} \mathcal{N}(\mathbf{0}, \Sigma)$ for all t .

A4. Known Σ .

Treating Σ as unknown and integrating it out under an Inverted-Wishart prior yields $\theta \mid \boldsymbol{\mu} \sim t_\nu(\tilde{\boldsymbol{\mu}}, \tilde{\sigma}^2)$ [Winkler, 1981]. Treating it as known retains normality but ignores estimation uncertainty. The feasibility of each route is very much a practical choice, as discussed in Section 1.2.

1.2 Precision Matrix Estimation

Proposition 1 makes clear that the entire framework depends on reliably obtaining the precision matrix Ω . In this section, we therefore turn to how Ω (or equivalently Σ) can be obtained. First of all, we note that the framework allows two practical routes in a forecasting application, for which we deliberately choose one over the other. The first route replaces Ω with an estimate, obtained via a frequentist or Bayesian method using an available data window, and plugs it back into Proposition 1 as if it were known. This route places no restriction on how Ω is estimated, so any estimation method can in principle be used. The second route instead treats Ω as unknown and integrates it out of the posterior entirely, which leads to a predictive t-distribution as in A4. This closed-form result, however, relies specifically on placing an Inverted Wishart prior on Σ , and does not extend to other choices of prior distribution.

In this thesis, we are interested in exploring estimation methods that extend some of the introduced assumptions: e.g., imposing sparsity, a factor structure, or allowing for non-stationarity. Since imposing these extensions requires a prior distribution other than the Inverted Wishart, we build on the first route. This route offers three advantages: (i) it places no restriction on how Ω is estimated, allowing for the extensions considered in this thesis, (ii) it allows Ω to be re-estimated separately for each data window, which is convenient in an empirical application, and (iii) working with a known Ω , rather than integrating Ω out analytically, means that no particular prior is required for a closed-form weights-solution to exist, which keeps computations

manageable even if $\mathbf{\Omega}$ must be re-estimated many times.

Now, before continuing further, we show a simple estimate for $\mathbf{\Omega}$ of both kinds, frequentist and Bayesian, to illustrate how one moves from forecast errors to a predictive distribution. The most basic frequentist estimate is the sample covariance matrix. Given T observations of the forecast error $\mathbf{u}_t$, the sample covariance matrix is defined as:

$$\hat{\mathbf{\Sigma}} = \frac{1}{T-1} \sum_{t=1}^T (\mathbf{u}_t - \bar{\mathbf{u}})(\mathbf{u}_t - \bar{\mathbf{u}})', \quad \bar{\mathbf{u}} = \frac{1}{T} \sum_{t=1}^T \mathbf{u}_t. \quad (1.3)$$

The estimate may be reasonably accurate if T is large relative to the number of forecasters p .

A Bayesian alternative is to estimate $\mathbf{\Sigma}$ by specifying a prior distribution. The natural-conjugate prior for the covariance matrix of a normal process is the Inverted Wishart distribution [Gelman et al., 2013]. The distribution is denoted as:

$$\mathbf{\Sigma} \sim IW(v, \mathbf{S}), \quad (1.4)$$

where v denotes the degrees of freedom and $\mathbf{S}$ is a scale matrix. Given T forecast errors, the posterior can be directly computed as:

$$\mathbf{\Sigma} \mid \mathbf{U} \sim IW(v + T, \mathbf{S} + \sum_{t=1}^T \mathbf{u}_t \mathbf{u}_t') \quad (1.5)$$

Intuitively, the prior can be seen as v pseudo-observations that move the posterior towards the scale matrix $\mathbf{S}$. The magnitude of v controls the confidence in $\mathbf{S}$. Since the Inverted Wishart distribution is only proper for $v > p + 1$, the degrees of freedom v must grow with the number of forecasters p , effectively requiring stronger prior confidence in $\mathbf{S}$. Conversely, when T becomes larger, the posterior converges towards the sample covariance. Importantly, updating the posterior with a new observation is as simple as adding $\mathbf{u}_t \mathbf{u}_t'$ to the scale matrix, without

the need for iterative sampling.

From estimates to a predictive distribution

Once an estimate of $\mathbf{\Omega}$ is available, there are two ways to turn it into a predictive distribution for θ . The simplest option is to plug a single point estimate $\hat{\mathbf{\Omega}}$ directly into Equation (1.2), which gives one predictive distribution $\mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$. This is cheap to compute, but treats $\hat{\mathbf{\Omega}}$ as if it were the truth: any uncertainty in estimating $\mathbf{\Omega}$ is ignored.

The Bayesian alternative is to place a prior on $\mathbf{\Omega}$ and obtain posterior draws $\mathbf{\Omega}^{(1)}, \dots, \mathbf{\Omega}^{(M)}$, for example via a Gibbs sampler. Each draw is plugged into Equation (1.2) separately, giving M predictive distributions $\mathcal{N}(\tilde{\mu}^{(m)}, \tilde{\sigma}^{2(m)})$. Averaging over these gives the overall predictive distribution for θ ,

$$p(\theta \mid \boldsymbol{\mu}) \approx \frac{1}{M} \sum_{m=1}^M \mathcal{N}\left(\theta; \tilde{\mu}^{(m)}, \tilde{\sigma}^{2(m)}\right). \quad (1.6)$$

This is referred to as Monte-Carlo integration: rather than solving the predictive distribution analytically, it is approximated numerically by averaging over posterior draws.¹ Every extension introduced in this thesis produces $\mathbf{\Omega}^{(1)}, \dots, \mathbf{\Omega}^{(M)}$ in a different way, but each is passed through Equation (1.6) in exactly the same way.

It should be noted that since $\boldsymbol{\mu}$ is fixed across draws, the posterior mean of θ can equivalently be written as $\bar{\mathbf{w}}' \boldsymbol{\mu}$, where $\bar{\mathbf{w}} = \frac{1}{M} \sum_m \mathbf{w}^{(m)}$ is the posterior mean weight vector. That is, rather than combining the M point forecasts $\tilde{\mu}^{(m)}$, we can equivalently average the M weight vectors first and apply the result to $\boldsymbol{\mu}$ once: $\bar{\mathbf{w}}$ plays exactly the role that $\mathbf{w}$ played in Proposition 1 under a known $\mathbf{\Omega}$.

¹If we were to draw $\mathbf{\Sigma}^{(1)}, \dots, \mathbf{\Sigma}^{(M)}$ from the Inverted Wishart posterior in Equation (1.5) and average the resulting predictive distributions as in Equation (1.6), the result converges, as $M \rightarrow \infty$, to the closed-form Student- t predictive distribution mentioned in A4. The Inverted Wishart case is thus a special case where Monte Carlo integration is unnecessary, since the exact answer it would converge to is already known analytically.

1.3 Extensions to the Winkler Framework

We now discuss how each of A1 - A4 is addressed in this thesis, starting with the assumption that drives most of the thesis' contribution. A4 (known Σ) is not 'strictly' relaxed in the sense of treating Σ as unknown and integrating it out analytically. Instead, this thesis interprets "known" pragmatically, in a way that permits different estimation methods to be substituted in its place: because Proposition 1 only requires a known Ω at the point where weights are computed, any estimate of Ω can play that role. Section 2.1 replaces the sample covariance estimate with a Bayesian graphical lasso prior, following Wang [2012], and derives the resulting posterior in detail. The prior shrinks many off-diagonal entries of Ω toward zero, effectively treating many pairs of forecasters as conditionally independent, which reduces the number of free parameters to estimate. Each entry is penalized independently, so that which pairs of forecasters are treated as conditionally independent is estimated directly from the data rather than fixed in advance. In principle, this could substantially improve the calibration and accuracy of the resulting forecast combination in a large panel of forecasters.

The graphical lasso prior is designed to improve on the sample covariance matrix, but it does so by imposing a specific structure on Ω that Proposition 1 itself does not require: sparsity. Whether this is the right structure to impose is not obvious. Two forecasters from different institutions, using different models, may have conditionally independent errors, which favors sparsity. In large panels of professional forecasters, however, errors may be driven by common macroeconomic shocks: GDP reports, inflation releases, central bank decisions. Sparsity alone cannot capture this. Section 2.2 addresses this with a factor graphical lasso: common variation is first absorbed by a number of latent factors, and sparsity is only imposed on the remaining forecaster-specific errors. Whether this structure actually improves the resulting forecast combination is an empirical question this thesis aims to examine.

The remaining assumptions are addressed more modestly, and independently of the extensions above. A1 (unbiasedness) is relaxed in Section 2.3: following Palm and Zellner [1992], forecasts are demeaned using the sample mean of past errors before any other extension is applied. A3 (stationarity) is relaxed in Section 2.4 through a discounting mechanism that downweights older

forecast errors, allowing Σ to vary over time. A2 (normality) is retained throughout.

2.1 Sparse Precision Estimation

Section 1.3 motivated the graphical lasso (GL) prior on $\mathbf{\Omega}$ as a way of reducing the number of free parameters to estimate, by shrinking many off-diagonal entries toward zero and treating the corresponding forecaster pairs as conditionally independent. It also avoids the constraint on the Inverted Wishart prior noted in Section 1.2, which forces an increasingly confident prior as the number of forecasters grows. This section develops the GL prior formally, following Wang [2012]: we define the prior, derive the posterior via a latent-variable representation, and construct the block Gibbs sampler used for posterior sampling. Readers that are only interested in the resulting algorithm can skip ahead to the sampling scheme at the end of this section.

The graphical lasso is a method for sparse precision matrix estimation and graph model determination. In a Gaussian graphical model, $w_{ij} = 0$ holds if and only if variables i and j are conditionally independent given all other variables. It is helpful to picture this as a graph: each forecaster is a node, and an edge is drawn between two forecasters when their errors are conditionally dependent. Estimating a sparse precision matrix $\mathbf{\Omega}$ essentially corresponds to erasing edges from this graph, one for every entry shrunk to exactly zero, until only the true dependencies remain. Friedman et al. [2008] propose to estimate such a sparse precision matrix by imposing a L_1 penalty on $\mathbf{\Omega}$. Let $\mathbf{S} = \mathbf{U}'\mathbf{U}$ be the cross-product of forecast errors. The objective is to maximize the penalized log-likelihood:

$$\max_{\mathbf{\Omega} \in M^+} \log \det \mathbf{\Omega} - \text{tr}(\mathbf{S}\mathbf{\Omega}) - \rho \|\mathbf{\Omega}\|_1 \quad (2.1)$$

where $\rho \geq 0$ is a shrinkage parameter, and $\|\mathbf{\Omega}\|_1 = \sum_{i,j} |\omega_{ij}|$ denotes the L_1 norm of $\mathbf{\Omega}$. The solution of Equation (2.1) yields a point estimate for $\mathbf{\Omega}$, with no quantification of estimation uncertainty. The GL prior provides a fully Bayesian counterpart.

Rather than directly optimising Equation (2.1), the Bayesian approach places a prior distribution on $\mathbf{\Omega}$. The GL prior on $\mathbf{\Omega}$ can be denoted as follows:

$$p(\mathbf{\Omega} \mid \lambda) \propto \prod_{i < j} \left\{ \frac{\lambda}{2} \exp(-\lambda |\omega_{ij}|) \right\} \prod_{i=1}^p \left\{ \frac{\lambda}{2} \exp\left(-\frac{\lambda}{2} \omega_{ii}\right) \right\} \mathbf{1}_{\mathbf{\Omega} \in M^+} \quad (2.2)$$

The prior consists of three important elements. First, the off-diagonal elements of the precision matrix w_{ij} are given a double exponential (Laplace) prior. The double exponential prior is symmetric and peaked at zero, shrinking partial correlations toward zero with strength determined by λ . Second, the diagonal elements of the precision matrix are given an exponential prior which ensures positive variance terms. Third, the indicator function $\mathbf{1}_{\mathbf{\Omega} \in M^+}$, which equals 1 if $\mathbf{\Omega} \in M^+$ and 0 otherwise, where M^+ is the space of $p \times p$ positive definite matrices, ensures $\mathbf{\Omega}$ is a valid precision matrix. It is noteworthy that, when combining the GL prior with the multivariate normal likelihood as in Equation (1.1), the posterior mode of $\mathbf{\Omega}$ coincides with the frequentist graphical lasso estimate that solves Equation (2.1), establishing a link between the Bayesian and frequentist approach to sparse precision estimation [Wang, 2012].

The GL prior on $\mathbf{\Omega}$ and the multivariate normal likelihood of Equation (1.1) together yield the posterior. Using Bayes' Theorem,

$$p(\mathbf{\Omega} \mid \mathbf{U}, \lambda) = \frac{p(\mathbf{U} \mid \mathbf{\Omega}) \cdot p(\mathbf{\Omega} \mid \lambda)}{p(\mathbf{U} \mid \lambda)} \quad (2.3)$$

where $p(\mathbf{U} \mid \lambda) = \int_{M^+} p(\mathbf{U} \mid \mathbf{\Omega}) \cdot p(\mathbf{\Omega} \mid \lambda) d\mathbf{\Omega}$ denotes the normalizing constant. Since $\mathbf{\Omega}$ must be positive definite by definition, the normalizing constant requires integrating over the space of positive definite matrices M_+ , for which no closed-form solution exists. Direct sampling from the posterior is thus intractable. A common trick in Bayesian statistics is to introduce some latent variable such that the posterior becomes tractable. Andrews and Mallows [1974] show that the double exponential distribution can be written as a scale mixture of normal distributions. If we introduce a latent scale parameter for each off-diagonal element $\boldsymbol{\tau} = \{\tau_{ij}\}_{i < j}$, the double exponential density in Equation (2.2) can be written as:

$$\frac{\lambda}{2} \exp(-\lambda |\omega_{ij}|) = \int_0^\infty \frac{1}{\sqrt{2\pi\tau_{ij}}} \exp\left(-\frac{\omega_{ij}^2}{2\tau_{ij}}\right) \cdot \frac{\lambda^2}{2} \exp\left(-\frac{\lambda^2}{2}\tau_{ij}\right) d\tau_{ij} \quad (2.4)$$

This representation allows us to write the GL prior in a hierarchical structure:

$$p(\boldsymbol{\omega} \mid \boldsymbol{\tau}, \lambda) = C_{\boldsymbol{\tau}}^{-1} \prod_{i < j} \left\{ \frac{1}{\sqrt{2\pi\tau_{ij}}} \exp \left(-\frac{\omega_{ij}^2}{2\tau_{ij}} \right) \right\} \prod_{i=1}^p \left\{ \frac{\lambda}{2} \exp \left(-\frac{\lambda}{2} \omega_{ii} \right) \right\} \mathbf{1}_{\boldsymbol{\Omega} \in M^+} \quad (2.5)$$

$$p(\boldsymbol{\tau} \mid \lambda) \propto C_{\boldsymbol{\tau}} \prod_{i < j} \frac{\lambda^2}{2} \exp \left(-\frac{\lambda^2}{2} \tau_{ij} \right) \quad (2.6)$$

where $C_{\boldsymbol{\tau}}$ is the analytically intractable normalizing constant of Equation (2.5). Equation (2.6) is cleverly constructed such that $C_{\boldsymbol{\tau}}$ cancels out when computing the joint distribution $p(\boldsymbol{\omega}, \boldsymbol{\tau} \mid \lambda) = p(\boldsymbol{\omega} \mid \boldsymbol{\tau}, \lambda) \cdot p(\boldsymbol{\tau} \mid \lambda)$, whilst still guaranteeing that $p(\boldsymbol{\tau} \mid \lambda)$ is proper [Wang, 2012]. Now crucially, marginalizing $p(\boldsymbol{\omega} \mid \lambda) = \int_0^\infty p(\boldsymbol{\omega}, \boldsymbol{\tau} \mid \lambda) d\boldsymbol{\tau}$ recovers the original GL prior of Equation (2.2).

We now turn to the full conditionals. Extending Equation (1.1) to the full sample of T observations $\mathbf{u}_1, \dots, \mathbf{u}_T$, the multivariate normal likelihood becomes:

$$p(\mathbf{U} \mid \boldsymbol{\Omega}) \propto |\boldsymbol{\Omega}|^{T/2} \exp \left\{ -\frac{1}{2} \text{tr}(\mathbf{S}\boldsymbol{\Omega}) \right\}, \quad (2.7)$$

where $\mathbf{S} = \mathbf{U}'\mathbf{U}$ denotes the cross-product of forecast errors. Combining Equation (2.7) with the hierarchical structure of Equations (2.5) - (2.6), we obtain the full posterior:

$$p(\boldsymbol{\Omega}, \boldsymbol{\tau} \mid \mathbf{U}, \lambda) \propto p(\mathbf{U} \mid \boldsymbol{\Omega}) \cdot p(\boldsymbol{\omega} \mid \boldsymbol{\tau}, \lambda) \cdot p(\boldsymbol{\tau} \mid \lambda), \quad (2.8)$$

which turns out to have tractable full conditional distributions $p(\boldsymbol{\Omega} \mid \boldsymbol{\tau}, \mathbf{U}, \lambda)$ and $p(\boldsymbol{\tau} \mid \boldsymbol{\Omega}, \mathbf{U}, \lambda)$, directly enabling a Gibbs sampler. Recall that we have only introduced $\boldsymbol{\tau}$ as a computational device to make the GL prior tractable, so that Gibbs samples of $\boldsymbol{\tau}$ can simply be discarded. We continue by deriving the full conditionals explicitly.

Gibbs Sampler

A natural starting point would be to update each element of $\mathbf{\Omega}$ one at a time, conditional on all other elements. However, this approach fails here since it can easily lead to a violation of the positive definiteness of the precision matrix ($\mathbf{\Omega} \in M^+$). Wang [2012] proposes a block update to deal with this issue, which implies updating an entire column (and, by symmetry, the corresponding row) of $\mathbf{\Omega}$ jointly at each step.

Without loss of generality, we focus on updating the last column and row of $\mathbf{\Omega}$. Given that the exact labeling of forecasters is arbitrary, the same derivation can be applied to any column once it has been moved to the last position. We partition $\mathbf{\Omega}$, the sample covariance matrix $\mathbf{S}$, and the matrix of latent scale parameters $\mathbf{\Upsilon} = \{\tau_{ij}\}$ as

$$\mathbf{\Omega} = \begin{pmatrix} \mathbf{\Omega}_{11} & \boldsymbol{\omega}_{12} \\ \boldsymbol{\omega}'_{12} & \omega_{22} \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} \mathbf{S}_{11} & \mathbf{s}_{12} \\ \mathbf{s}'_{12} & s_{22} \end{pmatrix}, \quad \mathbf{\Upsilon} = \begin{pmatrix} \boldsymbol{\Upsilon}_{11} & \boldsymbol{\tau}_{12} \\ \boldsymbol{\tau}'_{12} & 0 \end{pmatrix} \quad (2.9)$$

where $\mathbf{\Omega}_{11}$ is the $(p-1) \times (p-1)$ block obtained by removing the last row and column, and $\boldsymbol{\omega}_{12}$, ω_{22} are the remaining off-diagonal and diagonal elements.

To derive the full conditional of the last column, we isolate the terms in Equation (2.8) involving $\boldsymbol{\omega}_{12}$ and ω_{22} , and treat other terms as constants. First, we focus on Equation (2.7). We note that $|\mathbf{\Omega}| = |\mathbf{\Omega}_{11}| \cdot (\omega_{22} - \boldsymbol{\omega}'_{12} \mathbf{\Omega}_{11}^{-1} \boldsymbol{\omega}_{12}) \propto (\omega_{22} - \boldsymbol{\omega}'_{12} \mathbf{\Omega}_{11}^{-1} \boldsymbol{\omega}_{12})$ by the block-determinant identity, and $\text{tr}(\mathbf{S}\mathbf{\Omega}) = \text{tr}(\mathbf{S}_{11}\mathbf{\Omega}_{11}) + 2\mathbf{s}'_{12}\boldsymbol{\omega}_{12} + s_{22}\omega_{22}$, such that $\exp(-\frac{1}{2}\text{tr}(\mathbf{S}\mathbf{\Omega})) \propto \exp(-\frac{1}{2}(2\mathbf{s}'_{12}\boldsymbol{\omega}_{12} + s_{22}\omega_{22}))$. Next, we turn to the hierarchical GL prior in Equations (2.5) and (2.6). The off-diagonal factors of $p(\boldsymbol{\omega} \mid \boldsymbol{\tau}, \lambda)$ in Equation (2.5) involving the last column reduce to $\prod_{i < j=p} \exp\left(-\frac{\omega_{i,12}^2}{2\tau_{i,12}}\right) = \exp\left(-\frac{1}{2}\boldsymbol{\omega}'_{12}\mathbf{D}_{\boldsymbol{\tau}}^{-1}\boldsymbol{\omega}_{12}\right)$, where $\mathbf{D}_{\boldsymbol{\tau}} = \text{diag}(\boldsymbol{\tau}_{12})$. Finally, the diagonal factor of $p(\boldsymbol{\omega} \mid \boldsymbol{\tau}, \lambda)$ corresponding to ω_{22} is proportional to $\exp\left(-\frac{\lambda}{2}\omega_{22}\right)$, while $p(\boldsymbol{\tau} \mid \lambda)$ in Equation (2.6) does not depend on $\boldsymbol{\omega}_{12}$ or ω_{22} . Combining these four statements, the following conditional distribution of the last column arises:

$$p(\boldsymbol{\omega}_{12}, \omega_{22} \mid \boldsymbol{\Omega}_{11}, \boldsymbol{\Upsilon}, \mathbf{U}, \lambda) \propto (\omega_{22} - \boldsymbol{\omega}'_{12} \boldsymbol{\Omega}_{11}^{-1} \boldsymbol{\omega}_{12})^{T/2} \exp \left[-\frac{1}{2} \{ \boldsymbol{\omega}'_{12} \mathbf{D}_{\boldsymbol{\tau}}^{-1} \boldsymbol{\omega}_{12} + 2\mathbf{s}'_{12} \boldsymbol{\omega}_{12} + (s_{22} + \lambda) \omega_{22} \} \right]. \quad (2.10)$$

The above equation does not consist of a known kernel. Wang [2012] proposes the following change of variables:

$$\boldsymbol{\beta} = \boldsymbol{\omega}_{12}, \quad \gamma = \omega_{22} - \boldsymbol{\omega}'_{12} \boldsymbol{\Omega}_{11}^{-1} \boldsymbol{\omega}_{12} \quad (2.11)$$

Given that this substitution has a constant Jacobian, expanding $\omega_{22} = \gamma + \boldsymbol{\beta}' \boldsymbol{\Omega}_{11}^{-1} \boldsymbol{\beta}$ in the exponential and separately collecting terms in $\boldsymbol{\beta}$ and γ , we obtain:

$$p(\boldsymbol{\beta}, \gamma \mid \boldsymbol{\Omega}_{11}, \boldsymbol{\Upsilon}, \mathbf{U}, \lambda) \propto \gamma^{T/2} \exp \left(-\frac{s_{22} + \lambda}{2} \gamma \right) \times \exp \left(-\frac{1}{2} [\boldsymbol{\beta}' \{ \mathbf{D}_{\boldsymbol{\tau}}^{-1} + (s_{22} + \lambda) \boldsymbol{\Omega}_{11}^{-1} \} \boldsymbol{\beta} + 2\mathbf{s}'_{12} \boldsymbol{\beta}] \right), \quad (2.12)$$

which directly provides us two known kernels:

$$(\gamma, \boldsymbol{\beta}) \mid \boldsymbol{\Omega}_{11}, \boldsymbol{\Upsilon}, \mathbf{U}, \lambda \sim \text{Gamma} \left(\frac{T}{2} + 1, \frac{s_{22} + \lambda}{2} \right) \times N(-\mathbf{C} \mathbf{s}_{12}, \mathbf{C}), \quad \mathbf{C} = \{ \mathbf{D}_{\boldsymbol{\tau}}^{-1} + (s_{22} + \lambda) \boldsymbol{\Omega}_{11}^{-1} \}^{-1}, \quad (2.13)$$

where we have used that completing the square up to proportionality in Equation (2.12) yields $\exp(-\frac{1}{2}(\boldsymbol{\beta} + \mathbf{C} \mathbf{s}_{12})' \mathbf{C}^{-1}(\boldsymbol{\beta} + \mathbf{C} \mathbf{s}_{12}))$. A single Gibbs iteration for the last column (and row) of $\boldsymbol{\Omega}$ now simply involves sampling $(\gamma, \boldsymbol{\beta})$ from Equation (2.13) and mapping them back to $(\boldsymbol{\omega}_{12}, \omega_{22})$ using Equation (2.11).

We continue by showing that the above proposed block update preserves $\boldsymbol{\Omega} \in M^+$. Suppose that $\boldsymbol{\Omega}^{(t)}$ is positive definite at iteration t . Since the update leaves $\boldsymbol{\Omega}_{11}$ unchanged, it suffices to check the updated full matrix. By the same block-determinant identity as used before, $\det(\boldsymbol{\Omega}^{(t+1)}) = \gamma \cdot \det(\boldsymbol{\Omega}_{11}^{(t)})$, where $\det(\boldsymbol{\Omega}_{11}^{(t)}) > 0$ by assumption and $\gamma > 0$ by construction of

the Gamma distribution. Hence $\det(\mathbf{\Omega}^{(t+1)}) > 0$, so $\mathbf{\Omega}^{(t+1)}$ remains positive definite when using this Gibbs sampler.

The update for the latent scale parameters τ_{ij} is much simpler. Given that the likelihood does not involve $\boldsymbol{\tau}$, we only have to consider the prior terms of Equations (2.5) and (2.6), which yields:

$$p(\tau_{ij} \mid \mathbf{\Omega}, \lambda) \propto \tau_{ij}^{-1/2} \exp\left(-\frac{\omega_{ij}^2}{2\tau_{ij}} - \frac{\lambda^2}{2}\tau_{ij}\right), \quad i < j. \quad (2.14)$$

Applying the change of variables $u_{ij} = 1/\tau_{ij}$, each $1/\tau_{ij}$ is independently distributed as

$$1/\tau_{ij} \mid \mathbf{\Omega}, \lambda \sim \text{InvGauss}\left(\sqrt{\frac{\lambda^2}{\omega_{ij}^2}}, \lambda^2\right), \quad i < j. \quad (2.15)$$

Sampling Scheme

Having derived both full conditionals, the block Gibbs sampler cycles through the following steps at each iteration $t = 1, \dots, M$:

1. **Block update of $\mathbf{\Omega}$.** For each column $k = 1, \dots, p$, permute k to the last position, compute $\mathbf{C} = \{\mathbf{D}_{\boldsymbol{\tau}}^{-1} + (s_{22} + \lambda)\mathbf{\Omega}_{11}^{-1}\}^{-1}$, and sample

$$\gamma \sim \text{Gamma}\left(\frac{T}{2} + 1, \frac{s_{22} + \lambda}{2}\right), \quad \boldsymbol{\beta} \sim N(-\mathbf{C}\mathbf{s}_{12}, \mathbf{C}),$$

then recover $\omega_{12}^{(t)} = \boldsymbol{\beta}$ and $\omega_{22}^{(t)} = \gamma + \boldsymbol{\beta}'\mathbf{\Omega}_{11}^{-1}\boldsymbol{\beta}$.

2. **Update of $\boldsymbol{\tau}$.** For each $i < j$, sample

$$1/\tau_{ij}^{(t)} \sim \text{InvGauss}\left(\sqrt{\frac{\lambda^2}{\omega_{ij}^{(t)2}}}, \lambda^2\right).$$

After a burn-in of B iterations, the remaining $M - B$ draws $\{\mathbf{\Omega}^{(t)}\}_{t=B+1}^M$ provide us a sample from the posterior $p(\mathbf{\Omega} \mid \mathbf{U}, \lambda)$. Given that $\boldsymbol{\tau}$ was introduced only as a computational device,

those draws are discarded.

Adaptive Graphical Lasso

The double exponential prior in Equation (2.2) applies the same shrinkage λ to every off-diagonal entry. This is restrictive: a single λ tends to over-shrink large partial correlations and under-shrink small ones, since both are penalized at the same rate regardless of their size. The adaptive graphical lasso addresses this by letting each off-diagonal element carry its own shrinkage parameter λ_{ij} , so that the amount of regularization can respond to the size of the coefficient itself.

Following Wang [2012], we replace the single λ in Equation (2.2) with element-specific λ_{ij} :

$$p(\mathbf{\Omega} \mid \{\lambda_{ij}\}_{i < j}) \propto \prod_{i < j} \left\{ \frac{\lambda_{ij}}{2} \exp(-\lambda_{ij}|\omega_{ij}|) \right\} \prod_{i=1}^p \left\{ \frac{\lambda}{2} \exp\left(-\frac{\lambda}{2}\omega_{ii}\right) \right\} \mathbf{1}_{\mathbf{\Omega} \in M^+} \quad (2.16)$$

with an independent Gamma hyperprior $\lambda_{ij} \sim \text{Gamma}(r, s)$ for each $i < j$. The diagonal rate λ is left fixed, as in the standard GL prior. Model fitting proceeds with the same block Gibbs sampler described above, with one additional step: after updating $\mathbf{\Omega}$ and $\boldsymbol{\tau}$, each λ_{ij} is resampled from its full conditional.

Combining the λ_{ij} term in Equation (2.16) with its Gamma hyperprior provides a conjugate update:

$$\lambda_{ij} \mid \omega_{ij} \sim \text{Gamma}(1 + r, |\omega_{ij}| + s), \quad i < j. \quad (2.17)$$

This is the source of ‘adaptivity’: the conditional mean of λ_{ij} is $(1 + r)/(|\omega_{ij}| + s)$, which is large when ω_{ij} is small and small when ω_{ij} is large. Elements close to zero are pushed further toward zero, while large elements are shrunk less. The hyperparameters (r, s) need to be small so that λ_{ij} adapts to ω_{ij} rather than being controlled by the prior. We use $r = 10^{-2}$ and $s = 10^{-6}$, following the suggestion of Wang [2012].

2.2 Bayesian Factor Graphical Lasso

The graphical lasso prior described in Section (2.1) aims at estimating $\mathbf{\Omega}$, a sparse precision matrix of forecast errors. The approach can be expected to work well empirically if forecasters are conditionally independent. However, it is not obvious whether this is always the case in practice. For example, in a macro-economic setting, forecasters might respond to macroeconomic news in a very similar way, and so their errors might move together in ways that reflect a small number of common shocks. Directly placing sparsity on $\mathbf{\Omega}$ might not be the optimal way to capture this specific structure.

An alternative approach is to first remove the common variation shared across forecasters, and only then consider conditional independence between forecasters. We decompose the forecast errors as:

$$\mathbf{e}_t = \mathbf{\Lambda} \mathbf{f}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Omega}_{\varepsilon}^{-1}), \quad t = 1, \dots, T, \quad (2.18)$$

where $\mathbf{f}_t$ denotes a $q \times 1$ vector of unobservable factors with prior distribution $\mathbf{f}_t \sim N(\mathbf{0}, \mathbf{I}_q)$ independently across t , $\mathbf{\Lambda}$ denotes a $p \times q$ matrix of factor loadings and $\boldsymbol{\varepsilon}_t$ an idiosyncratic error term that is not explained by the common factors. The covariance matrix of forecast errors can now be decomposed as:

$$\mathbf{\Sigma} = \mathbf{\Lambda} \mathbf{\Lambda}' + \mathbf{\Omega}_{\varepsilon}^{-1} \quad (2.19)$$

After removing the common factors, it becomes much more defensible to assume that the remaining cross-forecaster dependence is sparse: the idiosyncratic errors of two forecasters are conditionally independent unless they share some institutional or methodological link not captured by the common factors. In other words, we could instead place the GL prior directly on $\mathbf{\Omega}_{\varepsilon}$.

[Lee and Seregina \[2025\]](#) also follow this approach, but estimate it using frequentist methods:

the factors are fixed by principal component analysis (PCA), and the frequentist graphical lasso discussed in Section (2.1) is applied to $\mathbf{\Omega}_\varepsilon$. The resulting point estimate $\hat{\mathbf{\Omega}}$ is plugged directly into the [Bates and Granger \[1969\]](#) weights formula, which is equivalent to the weights derived in Equation (1.2), resulting in combination weights with no associated measure of uncertainty. As before, treating $\mathbf{\Lambda}$ and $\mathbf{\Omega}_\varepsilon$ as unknown and estimating them with Bayesian inference restores this measure of uncertainty.

The approach proposed in this thesis therefore replaces both steps with Bayesian inference, following [Bhattacharya and Dunson \[2011\]](#), a standard paper for Bayesian factor modeling. However, their sampler is built for a diagonal idiosyncratic precision matrix $\mathbf{\Omega}_\varepsilon$, whilst the GL prior of Section 2.1 requires a general $\mathbf{\Omega}_\varepsilon$ with no diagonality restriction. Resolving this requirement is the main difficulty to overcome, and two strategies are available. The first strategy sidesteps the difficulty: it estimates $\mathbf{\Lambda}$ using the original diagonal sampler, and only afterwards applies the GL prior of Section 2.1 to residuals formed from a fixed estimate of $\mathbf{\Lambda}$. This sequential approach is straightforward to implement, but because $\mathbf{\Lambda}$ is fixed before the GL prior is applied, any posterior uncertainty in $\mathbf{\Lambda}$ does not propagate into the resulting draws of $\mathbf{\Omega}$. The second strategy, implemented in this thesis, addresses this difficulty: it samples $\mathbf{\Lambda}$ and $\mathbf{\Omega}_\varepsilon$ jointly within a single Gibbs sampler, so that uncertainty in the factor loadings and in the precision matrix are both reflected in the resulting predictive distribution.

Implementing the joint sampler requires two parts: a prior for the factor loadings $\mathbf{\Lambda}$, and a way to update this prior together with $\mathbf{\Omega}_\varepsilon$ within a single Gibbs sampler. The GL prior on $\mathbf{\Omega}_\varepsilon$ was already introduced in Section 2.1, so only the prior on $\mathbf{\Lambda}$ will be specified here. For posterior computation, we do not build a new sampler: we take [Bhattacharya and Dunson \[2011\]](#)'s existing Gibbs sampler and change the steps that assume a diagonal $\mathbf{\Omega}_\varepsilon$. We describe the prior on $\mathbf{\Lambda}$ first, and then turn to how the existing Gibbs sampler is modified to allow for joint sampling of $\mathbf{\Lambda}$ and $\mathbf{\Omega}_\varepsilon$.

Priors

To estimate $\mathbf{\Lambda}$ and select the number of factors q , we place a multiplicative gamma process (MGP) shrinkage prior on the factor loadings [Bhattacharya and Dunson, 2011]. Each entry of $\mathbf{\Lambda}$ is denoted λ_{jh} , the loading of forecaster j on factor h , for $j = 1, \dots, p$ and $h = 1, \dots, q$. Each entry λ_{jh} is given an independent normal prior,

$$\lambda_{jh} \mid \phi_{jh}, \tau_h \sim N(0, \phi_{jh}^{-1} \tau_h^{-1}), \quad \phi_{jh} \sim \text{Ga}(\nu/2, \nu/2), \quad (2.20)$$

with two components controlling its variance: a local parameter ϕ_{jh} specific to that one entry with prior mean 1, and a global parameter τ_h shared by the whole column h . The degrees of freedom ν controls how much ϕ_{jh} can vary around its mean, with $\nu = 3$ as default choice. The global parameter explains the name of the prior: τ_h is a running product of independent Gamma draws,

$$\tau_h = \prod_{l=1}^h \delta_l, \quad \delta_1 \sim \text{Ga}(a_1, 1), \quad \delta_l \sim \text{Ga}(a_2, 1) \text{ for } l \geq 2, \quad (2.21)$$

where each column's τ_h is the previous column's τ_{h-1} multiplied by one further Gamma draw. Under the restriction $a_2 > 1$, each δ_l is on average greater than one, so multiplying more and more of them together tends to push τ_h upward as h grows. Since τ_h enters the loading's precision, this means the prior variance shrinks as τ_h grows, so the degree of shrinkage increases with the column index of $\mathbf{\Lambda}$ on average: the first column of factor loadings is lightly shrunk, the second column a bit more, and so on. This process leads to later columns, corresponding to less important factors, being increasingly shrunk towards zero and effectively discarded. Intuitively, this shrinkage is what the sampler makes use of during posterior computation: it begins with a conservative upper bound on q and drops uninformative columns as the chain progresses, so that the data ends up determining how many factors are kept.

Posterior Computation

Two of Bhattacharya and Dunson [2011]'s six Gibbs sampler steps must be adapted to allow for the GL prior on $\mathbf{\Omega}_\varepsilon$: the factor loading update and the factor score update. We recommend the

reader to first read Section 3.1 of [Bhattacharya and Dunson \[2011\]](#), since we deliberately chose not to fully re-derive the entire algorithm here. The derivation below is written relative to their six steps, and is easiest to follow with their algorithm at hand.

The factor loadings can no longer be updated row by row. [Bhattacharya and Dunson \[2011\]](#)'s Step 1 relies on Σ_ε being diagonal, which makes the likelihood factorize across the p rows of $\mathbf{\Lambda}$. Once $\mathbf{\Omega}_\varepsilon$ has non-zero off-diagonal entries, the factorization no longer holds up, and all p rows of $\mathbf{\Lambda}$ must be updated jointly. Denoting the j th row of $\mathbf{\Lambda}$ by $\boldsymbol{\lambda}'_j$, and writing the log-likelihood as a quadratic form in $\boldsymbol{\lambda} = (\boldsymbol{\lambda}'_1, \dots, \boldsymbol{\lambda}'_p)'$, the vector stacking all p rows of $\mathbf{\Lambda}$, applying standard vectorization identities for Kronecker products yields (see Appendix A.2 for the full derivation):

$$\boldsymbol{\lambda} \mid \mathbf{F}, \mathbf{\Omega}_\varepsilon, \mathbf{U} \sim N_{pq}(\mathbf{Q}^{-1}\mathbf{b}, \mathbf{Q}^{-1}), \quad \mathbf{Q} = \mathbf{D} + \mathbf{\Omega}_\varepsilon \otimes (\mathbf{F}'\mathbf{F}), \quad \mathbf{b} = \text{vec}(\mathbf{F}'\mathbf{U}\mathbf{\Omega}_\varepsilon), \quad (2.22)$$

where $\mathbf{U}$ collects the T forecast errors $\mathbf{e}'_t$, $\mathbf{F}$ collects the T factor scores $\mathbf{f}'_t$, $\mathbf{D} = \text{blockdiag}(\mathbf{D}_1^{-1}, \dots, \mathbf{D}_p^{-1})$ with $\mathbf{D}_j^{-1} = \text{diag}(\phi_{j1}\tau_1, \dots, \phi_{jq}\tau_q)$ collects the row-wise MGP prior precisions of Equation (2.20), and $\text{vec}(\cdot)$ stacks the columns of its argument.

Equation (2.22) has the same structure as an ordinary Bayesian linear regression: the precision $\mathbf{Q}$ combines the prior precision $\mathbf{D}$ with a term reflecting how informative the data is about $\boldsymbol{\lambda}$, and the mean $\mathbf{Q}^{-1}\mathbf{b}$ shrinks the data-implied estimate towards zero, with the degree of shrinkage set by $\mathbf{D}$ relative to $\mathbf{\Omega}_\varepsilon \otimes (\mathbf{F}'\mathbf{F})$. The only complication is that $\boldsymbol{\lambda}$ stacks all p rows of $\mathbf{\Lambda}$ into one vector, since the rows can no longer be treated as separate regressions: $\mathbf{F}'\mathbf{F}$ appears exactly as it would in an ordinary regression, but $\mathbf{\Omega}_\varepsilon$ now also enters, since it controls how the p rows depend on each other. The Kronecker product $\mathbf{\Omega}_\varepsilon \otimes (\mathbf{F}'\mathbf{F})$ captures this: $\mathbf{F}'\mathbf{F}$ enters once per row, scaled by how strongly each pair of rows is linked through $\mathbf{\Omega}_\varepsilon$. Note that $\mathbf{Q}$ is $pq \times pq$, matching the length of $\boldsymbol{\lambda}$ as required. Equation (2.22) reduces to [Bhattacharya and Dunson \[2011\]](#)'s Step 1 exactly when $\mathbf{\Omega}_\varepsilon$ is diagonal, since $\mathbf{Q}$ then becomes block-diagonal and the p rows decouple.

Step 2, which samples each idiosyncratic precision σ_j^{-2} independently, is replaced entirely since

$\mathbf{\Omega}_\varepsilon$ is now updated as a full matrix via the Gibbs update of Section 2.1, applied to the residuals $\mathbf{E} = \mathbf{U} - \mathbf{F}\mathbf{\Lambda}'$. Now, the factor scores can be updated exactly as in [Bhattacharya and Dunson \[2011\]](#)'s Step 3, with $\mathbf{\Omega}_\varepsilon$ in place of their diagonal $\mathbf{\Sigma}_\varepsilon^{-1}$. Combining the $N(\mathbf{0}, \mathbf{I}_q)$ prior on $\mathbf{f}_t$ with the likelihood of $\mathbf{u}_t$ yields the conjugate update:

$$\mathbf{f}_t \mid \mathbf{\Lambda}, \mathbf{\Omega}_\varepsilon, \mathbf{u}_t \sim N_q \left((\mathbf{I}_q + \mathbf{\Lambda}'\mathbf{\Omega}_\varepsilon\mathbf{\Lambda})^{-1} \mathbf{\Lambda}'\mathbf{\Omega}_\varepsilon\mathbf{u}_t, (\mathbf{I}_q + \mathbf{\Lambda}'\mathbf{\Omega}_\varepsilon\mathbf{\Lambda})^{-1} \right), \quad t = 1, \dots, T. \quad (2.23)$$

The remaining updates now include the local shrinkage parameters ϕ_{jh} , the global shrinkage parameters τ_h , and the adaptive birth/death step that adjusts the number of retained columns q . These are all unchanged from [Bhattacharya and Dunson \[2011\]](#).

After burn-in, each retained draw $\{\mathbf{\Lambda}^{(m)}, \mathbf{\Omega}_\varepsilon^{(m)}\}$ is converted into a draw of Winkler's precision matrix via the Woodbury identity,

$$\mathbf{\Omega}^{(m)} = \mathbf{\Omega}_\varepsilon^{(m)} - \mathbf{\Omega}_\varepsilon^{(m)} \mathbf{\Lambda}^{(m)} \left(\mathbf{I}_q + \mathbf{\Lambda}^{(m)'} \mathbf{\Omega}_\varepsilon^{(m)} \mathbf{\Lambda}^{(m)} \right)^{-1} \mathbf{\Lambda}^{(m)'} \mathbf{\Omega}_\varepsilon^{(m)}. \quad (2.24)$$

The draws $\{\mathbf{\Omega}^{(m)}\}_{m=1}^M$ are then passed into the Winkler formula to obtain a predictive distribution.

2.3 Biased Forecasts

Assumption A1 (unbiasedness) is strong, and it is one that professional forecasters are documented to violate in practice: their errors often do not average to zero. We relax this by giving each forecaster i an unknown bias term θ_i , following [Palm and Zellner \[1992\]](#). Forecast errors are now

$$\mathbf{u} \sim \mathcal{N}(\boldsymbol{\theta}, \mathbf{\Sigma}), \quad \boldsymbol{\theta} = (\theta_1, \dots, \theta_p)'. \quad (2.25)$$

The bias is simply estimated by the sample mean of past forecast errors, $\hat{\theta}_i = T^{-1} \sum_{t=1}^T u_{it}$, and bias-corrected forecasts are defined as $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu}_{T+1} - \hat{\boldsymbol{\theta}}$: each forecaster's raw forecast is adjusted

by their own estimated bias before combination. Once this adjustment is made, $\tilde{\boldsymbol{\mu}}$ plays exactly the role that $\boldsymbol{\mu}$ played in Proposition 1, and the same result applies directly.

Proposition 2 (Palm & Zellner, 1992). *Under model (2.25), a diffuse prior on the target θ , and $\boldsymbol{\Omega}$ treated as known, the posterior for θ given the bias-corrected forecasts $\tilde{\boldsymbol{\mu}}$ is Gaussian with mean and variance*

$$\tilde{\mu}^* = \frac{\mathbf{e}'\boldsymbol{\Omega}\tilde{\boldsymbol{\mu}}}{\mathbf{e}'\boldsymbol{\Omega}\mathbf{e}}, \quad \tilde{\sigma}^{*2} = \frac{1}{\mathbf{e}'\boldsymbol{\Omega}\mathbf{e}}, \quad (2.26)$$

with weights $\mathbf{w}^* = \boldsymbol{\Omega}\mathbf{e}/\mathbf{e}'\boldsymbol{\Omega}\mathbf{e}$, identical to Winkler's, now applied to the bias-corrected forecasts.

Proof. Once bias has been removed, $\tilde{\boldsymbol{\mu}} = \theta\mathbf{e} + \mathbf{u}$ with $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$, which is exactly the setup of Proposition 1 with $\tilde{\boldsymbol{\mu}}$ in place of $\boldsymbol{\mu}$. The same completing-the-square argument yields (2.26). $\square$

The bias correction is a pre-processing step, entirely separate from the choice of precision estimator: any estimator of $\boldsymbol{\Omega}$ can be paired with it by simply working with $\tilde{\boldsymbol{\mu}}$ instead of $\boldsymbol{\mu}$. This separation is also a limitation. Correcting $\tilde{\boldsymbol{\mu}}$ does not change how $\boldsymbol{\Omega}$ is estimated: the GL sampler of Section 2.1 still runs on the raw forecast errors $\mathbf{u}$. A joint treatment would instead estimate θ and $\boldsymbol{\Omega}$ together, so that the uncertainty in estimating $\hat{\theta}$ is reflected in the posterior of $\boldsymbol{\Omega}$. We do not attempt this: it is a substantial extension to the GL sampler, and we leave it for future work.

2.4 Time-Varying Weights via Discounting

The sampling scheme in Section 2.1 relies on assumption A3: that forecast errors are identically distributed over time, so that $\boldsymbol{\Sigma}$ is stationary across the estimation window. This section relaxes A3 specifically for the Bayesian GL prior, allowing $\boldsymbol{\Sigma}$ to change gradually over time rather than staying fixed throughout the window. The Gibbs sampler in Section 2.1 uses assumption A3 because it takes the cross-product of forecast errors $\mathbf{S} = \mathbf{U}'\mathbf{U}$ as direct input in the sampling scheme. Since every observation enters this cross-product with the same weight, and the posterior for $\boldsymbol{\Omega}$ depends on the data only through $\mathbf{S}$, all past observations are treated as equally informative for the current precision matrix.

In practice, uncertain periods such as a financial crisis or a pandemic can cause the magnitude and correlation structure of forecast errors to change substantially, making older observations less informative for the current precision matrix [Rossi, 2013]. A simple fix is to replace $\mathbf{S}$ with a discounted version that places comparatively smaller weights on past observations, relaxing A3. Forecast errors are still treated as independent across time, but no longer as identically distributed within the window. A more involved but mathematically precise alternative would be the use of state space models with Kalman filtering [Raftery et al., 2010].

We define the discounted sample covariance matrix as:

$$\mathbf{S}_t(\delta) = \frac{1 - \delta}{1 - \delta^R} \sum_{s=0}^{R-1} \delta^s \mathbf{u}_{t-s} \mathbf{u}_{t-s}' \quad (2.27)$$

where R is the size of the estimation window and $\delta \in (0, 1)$ is the ‘forgetting factor’, which is named this way because an observation s periods in the past receives weight δ^s , decaying exponentially with s . The normalizing constant $\frac{1-\delta}{1-\delta^R}$ ensures the weights sum to exactly one for the finite window R . Note that this discounting is applied within the estimation window: relative to the rest of the window, the most recent quarter receives the largest weight, weights decay going backward through the window, and this pattern resets each time the window moves forward.

To obtain time-varying weights, we substitute $\mathbf{S}_t(\delta)$ for $\mathbf{S}$ in the Bayesian GL prior at each point in time and run the same Gibbs sampler as in Section 2.1 to obtain posterior draws of $\boldsymbol{\Omega}_t$, which are passed through Equation (1.2) exactly as before. Here, the sample size T from Section 2.1 corresponds to R , the size of the rolling estimation window. The Gibbs sampler’s conditional posteriors are derived assuming R independent, equally-weighted observations. However, discounting preserves independence but not equal weighting, so the effective number of independent observations shrinks below R . Writing $w_s = \delta^s / \sum_{k=0}^{R-1} \delta^k$ for the normalized discount weights, we take the effective sample size of the discounted window to be $R_{\text{eff}}(\delta) = 1 / \sum_{s=0}^{R-1} w_s^2 \leq R$. Left uncorrected, using R in the Gibbs sampler understates posterior uncertainty in $\boldsymbol{\Omega}_t$, so we substitute $R_{\text{eff}}(\delta)$ for R wherever the sampler’s conditional posteriors depend on sample size. Note in advance that Chapter 3 deliberately compares only this discounted specification against

the standard Bayesian GL, to isolate the effect of relaxing A3. Extending the correction to the other estimators is left for future work due to the added computational cost.

3 | Simulation Study

3.1 Design

Before evaluating any of the introduced methods from Chapter 2 on real data, it is useful to know how well they perform in a setting where the true data-generating process (DGP) is known. The goal of this simulation study is not necessarily to find a uniformly optimal method, but rather to see how our introduced methods perform under different data structures, and specifically under which circumstances it is actually beneficial to impose sparsity on $\mathbf{\Omega}$. The findings here also help interpret the empirical results in Chapter 4, where the true precision matrix is of course unknown.

The main difficulty in setting up a useful simulation is finding a DGP that is at once simple enough to be interpretable and rich enough to deliver some meaningful results. With this goal in mind, forecast errors for p forecasters at time t are drawn as

$$\mathbf{u}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}), \quad t = 1, \dots, T, \quad (3.1)$$

where $\mathbf{\Sigma}$ is a $p \times p$ block covariance matrix. The motivation for this ‘block’ covariance structure is that, in a panel of forecasters, it is natural to assume that some forecasters share an institutional background, similar models, or belong to a similar sector and therefore make correlated errors, whereas forecasters from different groups are only weakly correlated. A block covariance structure captures this: it allows for strong within-group dependence and weaker cross-group dependence, and it lets us control sparsity directly. If cross-group dependence is zero, the blocks are independent by construction and $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$ is exactly block-diagonal, which gives a perfectly sparse precision matrix. As cross-group dependence increases, $\mathbf{\Omega}$ becomes increasingly dense. We do not claim that the multivariate normal DGP with this covariance structure is fully realistic: it is chosen because it gives a transparent setting in which the true degree of sparsity in $\mathbf{\Omega}$ is known and can be varied directly.

We divide the forecasters into K groups of unequal size. Within-group correlation is fixed at

$\rho_{within} = 0.8$ and cross-group correlation is fixed at $\rho_{across} \in \{0.0, 0.1, 0.3\}$. To provide some intuition, consider a specific example with $p = 5$ forecasters divided into $K = 2$ blocks of size 2 and 3. Setting $\rho_{within} = 0.8$ and $\rho_{across} = 0.1$, the covariance matrix of the DGP takes the form:

$$\mathbf{\Sigma} = \begin{pmatrix} 1 & 0.8 & 0.1 & 0.1 & 0.1 \\ 0.8 & 1 & 0.1 & 0.1 & 0.1 \\ 0.1 & 0.1 & 1 & 0.8 & 0.8 \\ 0.1 & 0.1 & 0.8 & 1 & 0.8 \\ 0.1 & 0.1 & 0.8 & 0.8 & 1 \end{pmatrix}.$$

Before continuing any further, it is useful to get some intuition into how the structure of $\mathbf{\Sigma}$ shapes the Winkler weights themselves. In the above example, two features of $\mathbf{\Sigma}$ stand out. The first is block size, where each forecaster in the smaller block (size 2) carries more weight individually than those in the larger block (size 3), since there are fewer of them to split the group's weight between. This is visible in Table 3.1 even at $\rho_{across} = 0.0$, where the blocks are entirely independent: each forecaster in block 1 gets a weight of 0.245, against 0.170 for each forecaster in block 2. The second feature is ρ_{across} , which controls how the two blocks relate to one another. As Table 3.1 shows, moving from $\rho_{across} = 0.0$ to 0.3 does not affect the weights much, but pushing ρ_{across} closer to $\rho_{within} = 0.8$ pulls the weights toward the equal-weighted average of $1/5 = 0.2$, since at that point the block structure has basically disappeared. This toy example provides a simple takeaway: it is the block structure of $\mathbf{\Omega}$ that drives unequal weighting across groups, and in denser matrices, higher cross-block correlation pushes the weights closer to one another. Note that this example uses the true $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$ directly, whereas the simulation below estimates $\mathbf{\Omega}$ from simulated error terms, so this mechanism should be seen in isolation from the estimation noise that follows.

ρ_{across}	group 1 ($n = 2$) total weight	group 2 ($n = 3$) total weight	per-forecaster, group 1	per-forecaster, group 2
0.0	0.491	0.509	0.245	0.170
0.1	0.489	0.511	0.245	0.170
0.3	0.486	0.514	0.243	0.171
0.7	0.455	0.545	0.227	0.182

Table 3.1: Implied Winkler weights $w = \mathbf{\Omega e} / \mathbf{e' \Omega e}$ for the $p = 5$, $K = 2$ toy example, computed from $\mathbf{\Omega} = \mathbf{\Sigma}^{-1}$ at $\rho_{\text{within}} = 0.8$ and varying ρ_{across} (the $\rho_{\text{across}} = 0.1$ row corresponds exactly to the $\mathbf{\Sigma}$ shown above). The $\rho_{\text{across}} = 0.7$ row is included only to illustrate the convergence towards equal weighting and does not correspond to a value used in the simulation grid.

In the simulation, block sizes are drawn once per (p, K) combination by normalizing K independent Exponential(1) draws to sum to p and rounding to integers, then fixed across all replications. Besides ρ_{across} , the number of blocks K also affects sparsity, even at $\rho_{\text{across}} = 0$: with more, smaller blocks, a larger fraction of the $\binom{p}{2}$ off-diagonal entries of $\mathbf{\Omega}$ are exactly zero, since the total number of within-block pairs falls as p is split into more groups. Holding p fixed, moving from $K = 2$ to $K = 5$ therefore increases the true sparsity level even before ρ_{across} is changed.

Parameter	Values
Forecasters p	5, 10, 20, 30, 50
Blocks K	2, 3, 5
Cross-block correlation ρ_{across}	0.0, 0.1, 0.3
Sample size T	100
Replications	20
Within-block correlation ρ_{within}	0.8 (fixed)

Table 3.2: Simulation grid parameters.

The full simulation grid is summarized in Table 3.2. We compare five methods for estimating $\mathbf{\Omega}$: two baselines (the sample covariance inverse and the Inverted Wishart posterior) and the three methods introduced in Chapter 2 (the Bayesian GL, the adaptive Bayesian GL, and the Bayesian factor GL). Four of the five methods require posterior sampling; the sample covariance inverse does not, since it is a single plug-in estimate. The Inverted Wishart posterior is conjugate and drawn from directly, so no burn-in is needed. The Bayesian GL, adaptive Bayesian GL, and Bayesian factor GL are all run for $M = 1000$ post-burn-in draws, with a burn-in of 500 for the Bayesian GL and Bayesian factor GL and 1000 for the adaptive Bayesian GL. We picked the

longer burn-in for the adaptive method a priori, since its per-edge shrinkage seemed like it might need more iterations to settle. Chapter 4 carries out a detailed burn-in check on real forecasting data, since that is where slow mixing and convergence turned out to matter. Here we only ran a smaller version of the same check, since the synthetic data is much simpler. We looked at trace plots of $\mathbf{\Omega}$'s diagonal and a cross-block entry at $p = 20$ and at $p = 50, \rho_{\text{across}} = 0.3$ (the densest cell in the grid). All three samplers looked stable well before either burn-in point. The joint sampler of Section 2.2 is initialized at $k_{\text{init}} = \lfloor 3 \log p \rfloor$ factors with no upper limit on k . This is a deliberate difference from the empirical application in Chapter 4, where k is capped on economic grounds, and is worth keeping in mind when interpreting the results below.

The sample covariance inverse is the only method without a posterior: it plugs a single point estimate $\hat{\mathbf{\Omega}}$ directly into the Winkler weights formula of Equation (1.2), following the plug-in route described in Section 1.2. The remaining four methods follow the Monte Carlo integration route instead, producing M posterior draws $\mathbf{\Omega}^{(1)}, \dots, \mathbf{\Omega}^{(M)}$ that are each passed through the weights formula and combined via Monte Carlo integration.

Two evaluation criteria are used. The primary criterion is the mean squared error (MSE) of the implied Winkler weights, evaluated per replication as

$$\text{MSE} = \frac{1}{M} \sum_{m=1}^M \|w^{(m)} - w_{\text{true}}\|_2^2, \quad (3.2)$$

where $w^{(m)} = \mathbf{e}'\mathbf{\Omega}^{(m)}/\mathbf{e}'\mathbf{\Omega}^{(m)}\mathbf{e}$, and w_{true} is obtained the same way but using the known $\mathbf{\Omega}_{\text{true}} = \mathbf{\Sigma}_{\text{true}}^{-1}$ in place of $\mathbf{\Omega}^{(m)}$, since $\mathbf{\Sigma}_{\text{true}}$ is known by construction in the simulation. For the sample covariance inverse, $M = 1$ and the formula reduces to a single squared-error term. The secondary criterion is the log score. The log score uses the predictive density from Equation (1.2), evaluated at a single test-period forecast vector $\mu_{\text{test}} = \theta_{\text{test}} + u_{\text{test}}$, with $\theta_{\text{test}} \sim N(0, 1)$ and $u_{\text{test}} \sim N(0, \mathbf{\Sigma})$ drawn from the same $\mathbf{\Sigma}$ used to generate that replication's training data. Both θ_{test} and μ_{test} are fixed for the replication; only the posterior draws $\mathbf{\Omega}^{(m)}$, and hence $\tilde{\mu}^{(m)}$ and $\bar{\sigma}^{2(m)}$, vary

across the sum below. The log score is then

$$\text{LS} = \log \left(\frac{1}{M} \sum_{m=1}^M \mathcal{N}(\theta_{\text{test}}; \tilde{\mu}^{(m)}, \tilde{\sigma}^{2(m)}) \right), \quad (3.3)$$

evaluated at the true value θ_{test} .¹ Only one $(\theta_{\text{test}}, u_{\text{test}})$ pair is drawn per replication, which aligns with reality: at each point in time, there is only one true value to forecast. This is why the log score is noisier across replications than the MSE.

3.2 Results

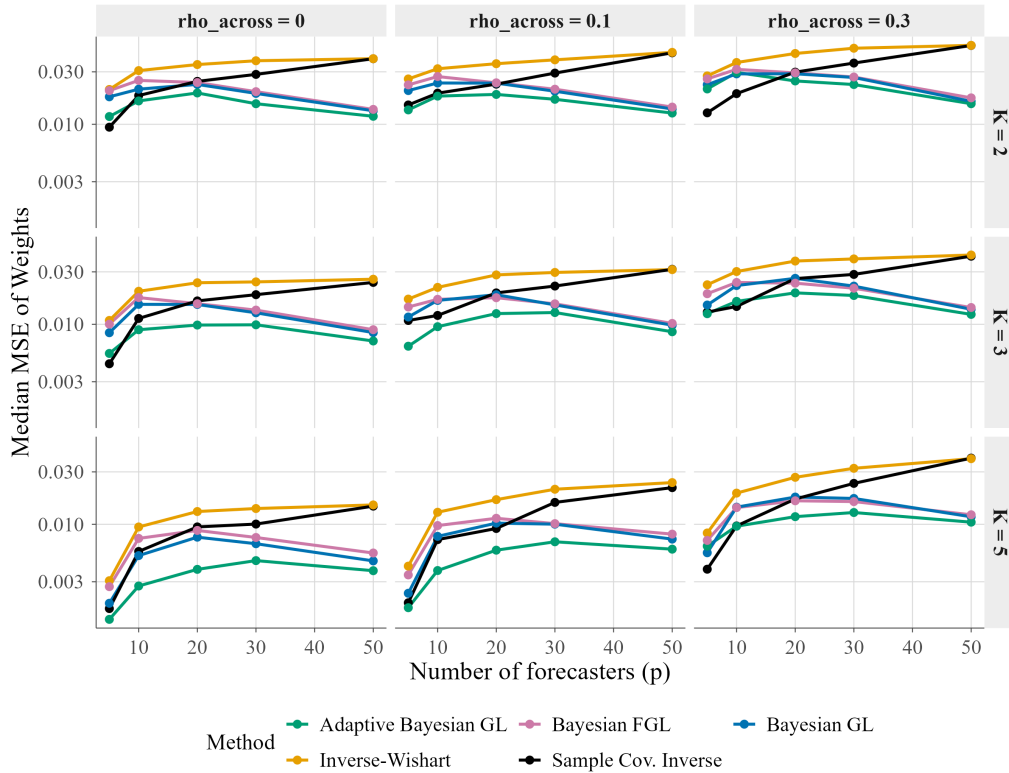


Figure 3.1: Median MSE of implied Winkler weights across the simulation grid. Rows correspond to the number of blocks $K \in \{2, 3, 5\}$ and columns to cross-block correlation $\rho_{\text{across}} \in \{0.0, 0.1, 0.3\}$. Within-block correlation is fixed at $\rho_{\text{within}} = 0.8$ and sample size at $T = 100$. Points show the median across 20 replications, the y -axis is on a log scale.

¹The value drawn for θ_{test} does not affect the log score. Shifting θ_{test} shifts μ_{test} by the same amount, and thus $\tilde{\mu}^{(m)}$ too, leaving the gap $\tilde{\mu}^{(m)} - \theta_{\text{test}}$ unchanged.

Figure 3.1 reports the median MSE of the implied Winkler weights across the simulation grid. The rows correspond to $K \in \{2, 3, 5\}$ and columns to $\rho_{\text{across}} \in \{0.0, 0.1, 0.3\}$.

The most consistent finding is that the GL-based methods achieve substantially lower median MSE compared to the other methods across most of the grid cells, with the difference growing as the number of forecasters p increase towards $T = 100$. The result matches expectation: as the number of forecasters grows, the sample covariance inverse becomes an increasingly unreliable estimate of $\mathbf{\Omega}$, and the sparsity induced by the GL prior makes estimation of the covariance matrix more robust, so that the implied Winkler weights are closer to the truth. Furthermore, the adaptive Bayesian GL consistently provides a small edge over the Bayesian GL and Bayesian factor GL at a high number of forecasters, although the difference is small compared to the gap with the unregularized methods.

Conversely, the Inverted Wishart prior and the sample covariance estimator start performing consistently worse than the three GL-based methods as the number of forecasters grows. The sample covariance inverse offers a marginal improvement over the Inverted Wishart prior, since neither places any weight on sparsity: the Inverted Wishart prior shrinks the estimate towards a scale matrix but does not favour a sparse graph structure, so the block pattern in $\mathbf{\Omega}$ is not recovered any more effectively than by directly inverting the sample covariance matrix.

The Bayesian factor GL consistently tracks slightly behind the Bayesian GL and Adaptive Bayesian GL as the number of forecasters grows. Its median MSE follows a single-peaked shape in p , rising then falling, a pattern that recurs across almost every grid cell. First, the DGP does not impose a factor structure: the block covariance is not the product of a small number of common factors plus idiosyncratic noise, so the factor-extraction step has nothing to remove and instead produces noise that is then passed to the graphical lasso step. A second thing to note is that k is left uncapped in this simulation, unlike the empirical application in Chapter 4. In the joint sampler, the factor loadings and $\mathbf{\Omega}_\varepsilon$ are updated conditional on each other at each iteration, so any spurious structure the factor step picks up from noise (given that no real factor structure exists here) feeds into $\mathbf{\Omega}_\varepsilon$ and back into the next iteration's factor update, with nothing

to stop k from drifting upward in response.² Overall, there is a clear mismatch between model and data, and it suggests the method’s relative standing may improve further in an empirical application, where any latent factor structure, if it exists, is not ruled out by construction.

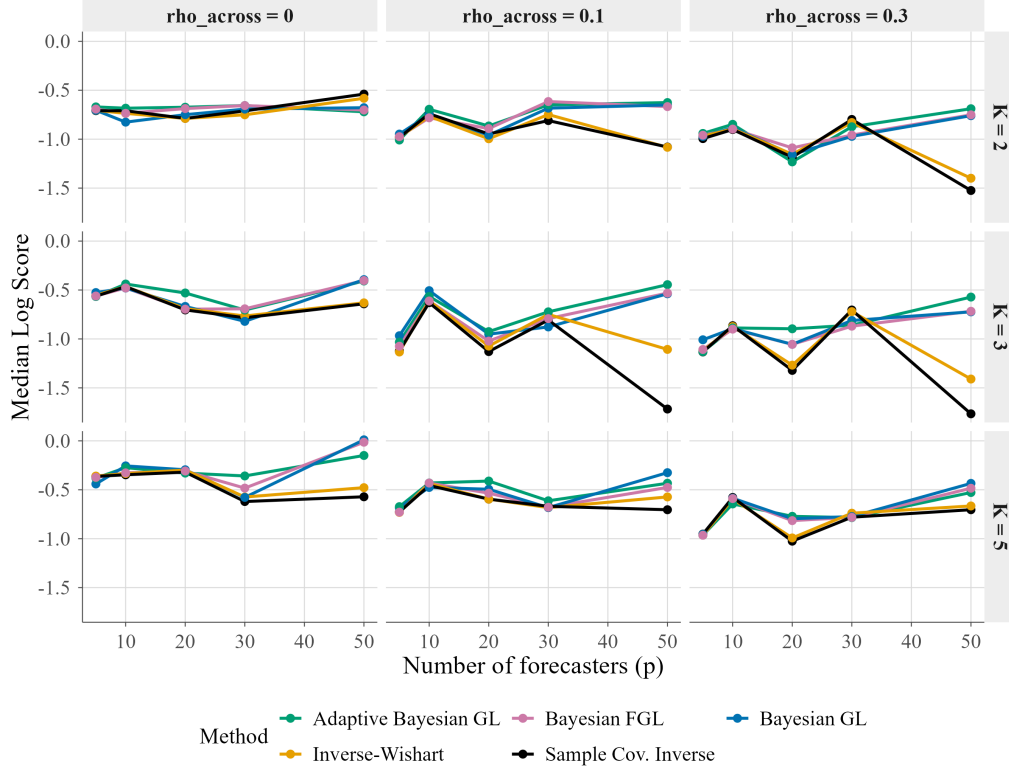


Figure 3.2: Median log score across the simulation grid, using the same row/column structure as Figure 3.1.

The log score reported in Figure 3.2 is considerably noisier than the MSE, and its ranking across methods is less consistent. This is expected: unlike the MSE, which already averages over all M posterior draws within a replication, the log score is evaluated at a single held-out test point per replication. Looking at the underlying replication-level spread confirms this (not shown in Figure 3.2, since only medians are plotted to keep it clean): the variability in log score is wide enough that the methods’ typical ranges overlap substantially, whereas for the MSE this overlap is much more limited. As a result, the method with the best median log score in a given cell frequently differs from the method with the best median MSE. Still, the GL-based methods

²We investigate this factor-drifting phenomenon more closely in the empirical chapter.

provide improved density calibration at $p = 30$ and $p = 50$ across most of the simulation grid.

Robustness to bias

As a final check, we investigate what happens when the zero-mean assumption is violated and the estimation methods do not correct for it. As stated in Equation (1.1), Winkler’s weights formula assumes that forecast errors average to zero, which is why the methods in this thesis do not center the data in advance. If forecasters are actually biased, this can change results in two ways. First, an unmodelled bias can worsen the estimate of $\mathbf{\Omega}$ itself, since the true error covariance matrix is then mixed with a rank-one bias term.³ Second, even if $\mathbf{\Omega}$ were estimated perfectly, bias can still distort the final forecast directly, since the Winkler weights are just applied to whatever forecasts they are given, biased or not. Our simulation excludes the bias correction of Chapter 2, where each forecaster’s bias is estimated and removed from the predictive mean. Here, no such correction is applied, so we can see directly how an unmodeled bias affects results.

To implement this, a bias vector $\mathbf{b} = (b_1, \dots, b_p)'$ is drawn once per replication, with each $b_i \sim N(0, \tau^2)$ drawn independently across forecasters. This same $\mathbf{b}$ is added to every training-period error and to the test-period forecast within that replication. None of the five methods are ‘informed’ about $\mathbf{b}$: they are simply run on data that no longer satisfies Winkler’s zero-mean assumption.

³The sample covariance is a special case here: it computes $\frac{1}{T-1} \sum_t (\mathbf{u}_t - \bar{\mathbf{u}})(\mathbf{u}_t - \bar{\mathbf{u}})'$ by definition, and so always subtracts off the sample mean regardless of bias, which implies it is ‘corrected’ for the first source through which bias enters. However, it is clearly not robust to the second.

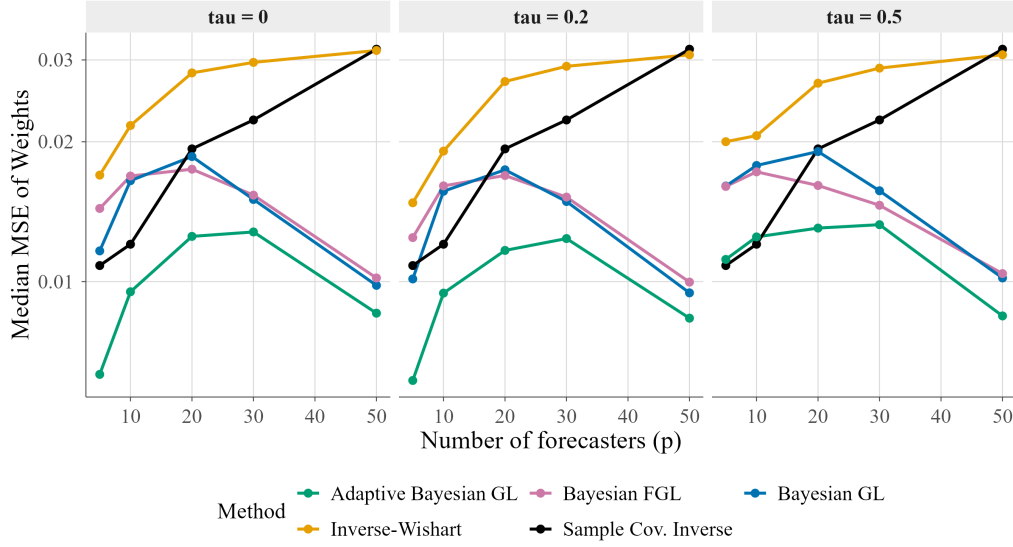


Figure 3.3: Median MSE of implied Winkler weights under non-zero forecaster bias, for $K = 3$ and $\rho_{\text{across}} = 0.1$. Panels correspond to $\tau \in \{0, 0.2, 0.5\}$, the standard deviation of the forecaster-specific bias added to each series. $\tau = 0$ reproduces the corresponding cell from Figure 3.1.

Figure 3.3 reports results for $\tau \in \{0, 0.2, 0.5\}$. We fix $K = 3$ and $\rho_{\text{across}} = 0.1$, a middle case within the main grid, since extending the bias check to the full grid would add considerably to computational cost without changing our conclusions. The MSE ranking from the main grid is mostly preserved, and the added bias term mainly changes the simulation results at small p . Between $\tau = 0$ and $\tau = 0.5$, MSE rises sharply at $p = 5$, by roughly 80% for the Adaptive Bayesian GL and by roughly 40% for the Bayesian GL, while at $p = 50$ both are barely affected. Clearly, the fixed bias term contributes less to the total estimation error as the number of forecasters p grows. It can also be noted that the sample covariance inverse outperforms the Inverted Wishart prior, which is a direct consequence of what was earlier described in the footnote. This advantage is however less meaningful than it appears: Equation (3.2) measures MSE on the estimated weights $w^{(m)}$ themselves, not on the resulting forecasts, and in practice the same Winkler weights are still applied to a biased forecast vector regardless of which method produced them. The sample covariance inverse’s ‘improvement’ on this particular metric therefore does not imply better forecasts under bias, only a more stable estimate of Ω .⁴

⁴Developing Bayesian precision matrix estimators that are robust to forecaster bias is non-trivial and is left for future work, as it falls outside the scope of this thesis.

The log score adds a few interesting insights: at $p = 50$, the sample covariance inverse's median log score falls from about -1.7 at $\tau = 0$ to about -2.3 at $\tau = 0.5$, whilst the Bayesian GL, Adaptive Bayesian GL and Bayesian factor GL stay essentially flat around -0.4 to -0.5 regardless of τ . So a sparse or shrunk estimate of $\mathbf{\Omega}$ does not fix biased weights, but it does keep the predictive density stable under high forecaster dimensions, unlike the sample covariance inverse.

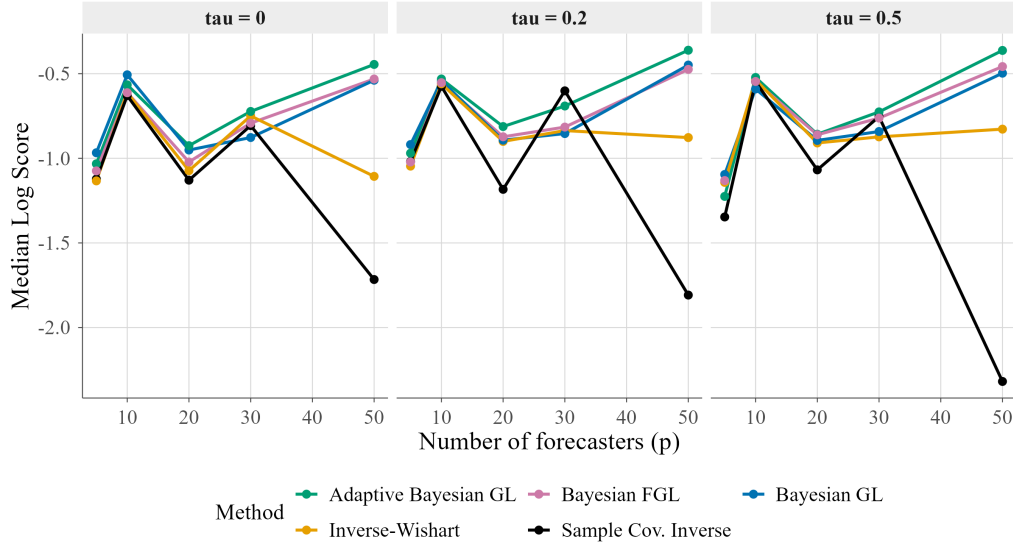


Figure 3.4: Median log score under non-zero forecaster bias, for $K = 3$ and $\rho_{\text{across}} = 0.1$. Panels correspond to $\tau \in \{0, 0.2, 0.5\}$, the standard deviation of the forecaster-specific bias added to each series. $\tau = 0$ reproduces the corresponding cell from Figure 3.2.

4 | Empirical Application

This chapter applies the framework and methods developed over Chapters 2 and 3 to a real forecasting panel: quarterly GDP growth forecasts from the ECB Survey of Professional Forecasters. Unlike the simulation study, the true precision matrix is unknown here, so we cannot measure how close each method gets to the truth directly. Instead, we evaluate forecasting performance out-of-sample using the realized GDP growth for each quarter, published after the forecasts are made. The chapter first describes the data and rolling-window setup used to generate out-of-sample forecasts, then compares the six methods against each other, and finally discusses three modeling choices: the EWMA ‘forgetting’ factor, the factor cap in Bayesian factor GL, and the burn-in length for each MCMC method.

4.1 Data and Estimation

The empirical application uses point forecasts of euro area real GDP growth from the ECB Survey of Professional Forecasters (SPF), covering 1999Q3–2026Q1 ($T = 107$ quarterly observations) with $p = 35$ forecasters reporting over this period. Realized values are taken from Eurostat’s national accounts release. Forecast errors are computed as $u_{it} = y_t - f_{it}$, where f_{it} is forecaster i ’s point forecast for quarter t and y_t the realized value. All series are rescaled to percentage points before estimation.

The panel of 35 forecasters is obtained following the filtering approach of [Lee and Seregina \[2025\]](#). The raw SPF release counts over a hundred individual forecasters, but participation is highly irregular: many respondents enter or exit the panel throughout the sample, leaving a large number of gaps. We drop any forecaster missing more than 40% of rounds, which removes irregular reporters whilst retaining a useful panel size. The remaining gaps are filled using the random forest imputation algorithm of [Stekhoven and Bühlmann \[2012\]](#). This leaves $p = 35$ forecasters with a complete panel over $T = 107$ quarters. The panel size matters for this thesis: sparsity in Ω is only worth studying if p is large enough to make precision estimation difficult, so keeping the panel size substantial is a deliberate choice, not a data-cleaning detail. [Figure 4.1](#)

shows the resulting panel after filtering and imputation.

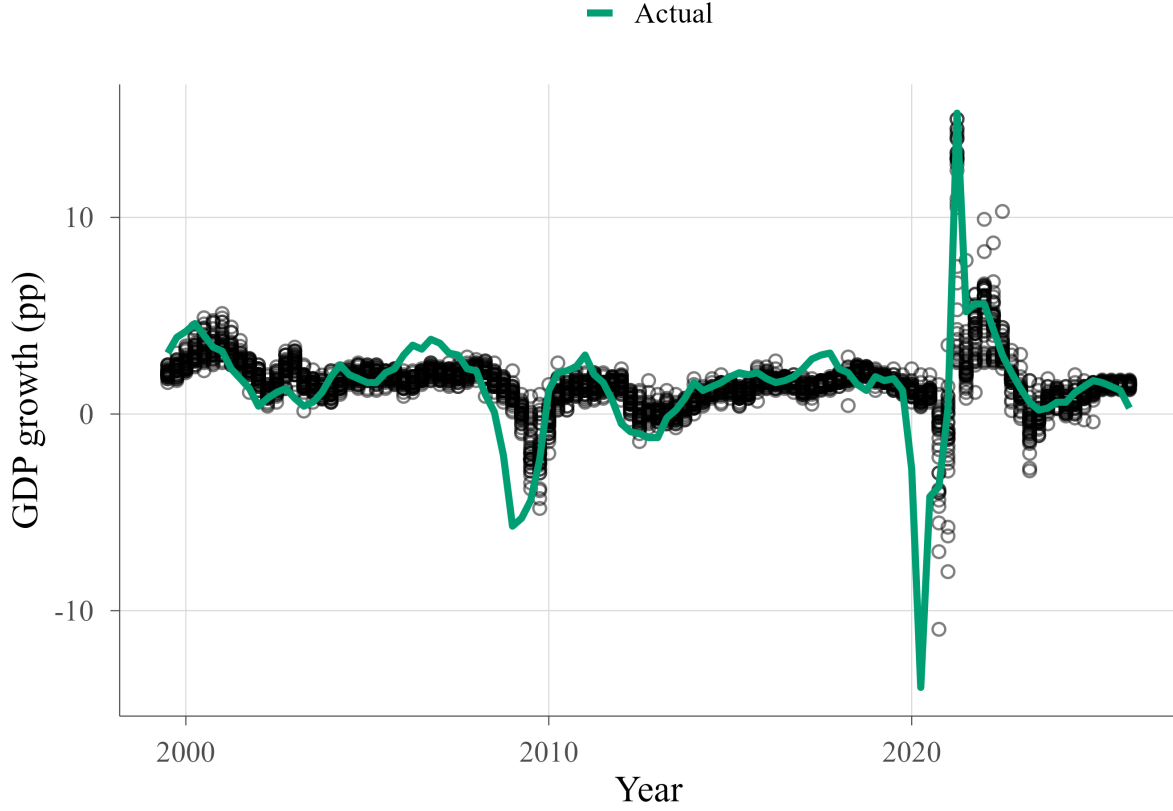


Figure 4.1: Point forecasts of euro area GDP growth from the ECB SPF panel (circles), one quarter ahead, against realized GDP growth (green line), 1999Q3–2026Q1.

We evaluate the framework out-of-sample using a rolling window of size R : at each period t , the previous R quarters of forecast errors are used to estimate $\mathbf{\Omega}$ and, where applicable, the bias vector $\hat{\boldsymbol{\theta}}$. The resulting Winkler weights are applied to the forecasts submitted for quarter $t + 1$. We report results for $R \in \{50, 60, 70\}$, yielding 57, 47, and 37 test periods respectively. R is chosen well above $p = 35$, so that the sample covariance matrix remains identified.

Six methods for estimating $\mathbf{\Omega}$ are compared: the sample covariance inverse, the Inverted Wishart prior, the Bayesian GL, the adaptive GL, the EWMA-discounted GL, and the Bayesian factor GL. All methods except the sample covariance inverse produce $M = 1000$ post-burn-in posterior draws, which are passed through the Winkler weights formula and combined via Monte Carlo

integration, as in the simulation study. All methods are evaluated with and without the bias correction of Section 2.3, reusing the same posterior draws for both.

4.2 Results

Rolling-window evaluation and the two metrics

At each test period t , the estimation window covers the R quarters preceding t , $\mathbf{U}_{t-R:t-1}$. This window provides us the bias estimate $\hat{\boldsymbol{\theta}}_t = \frac{1}{R} \sum_{s=t-R}^{t-1} \mathbf{u}_s$ and the posterior draws $\boldsymbol{\Omega}_t^{(1)}, \dots, \boldsymbol{\Omega}_t^{(M)}$. Neither uses information from period t or later, so every result below is out-of-sample. The window then rolls forward one quarter and the procedure repeats.

At t , each draw $\boldsymbol{\Omega}_t^{(m)}$ yields a predictive mean and variance $(\tilde{\mu}_t^{(m)}, \tilde{\sigma}_t^{2(m)})$ via Equation (1.2), computed once using the raw forecasts $\mathbf{f}_t$ and once using the bias-corrected forecasts $\tilde{\mathbf{f}}_t = \mathbf{f}_t - \hat{\boldsymbol{\theta}}_t$, so that both versions are evaluated from the same posterior draws. Two metrics are computed from the resulting M draws, following the Monte Carlo integration route of Equation (1.6). The squared forecast error is averaged over posterior draws first,

$$\text{SFE}_t = \frac{1}{M} \sum_{m=1}^M \left(y_t - \tilde{\mu}_t^{(m)} \right)^2, \quad (4.1)$$

so that by averaging the M draws, we treat each as a separate prediction, rather than collapsing them into a single point forecast before scoring them. The log score is built the same way at the level of densities,

$$\text{LS}_t = \log \left(\frac{1}{M} \sum_{m=1}^M \mathcal{N} \left(y_t; \tilde{\mu}_t^{(m)}, \tilde{\sigma}_t^{2(m)} \right) \right). \quad (4.2)$$

Both metrics are computed identically for the raw and bias-corrected forecasts, and both are then averaged across all $T - R$ test periods to give the MSFE and mean log score reported below. For the sample covariance inverse, which gives a single point estimate $\hat{\boldsymbol{\Omega}}$ rather than a posterior, $M = 1$ and both formulas reduce to the ordinary squared error and log density at a single estimate, matching the simulation study.

Headline comparison

Method	$R = 50$	$R = 60$	$R = 70$
Sample Cov. Inverse	1.000 (0.012)	1.000 (0.004)	1.000 (0.004)
Inv-Wishart	0.237	0.823	0.795
Bayes GL	0.209	0.602	0.648
Adaptive GL	0.137	0.411	0.573
EWMA	0.344	0.858	0.895
BFGL	0.223	0.626	0.609

Table 4.1: Ratio of each method’s MSFE relative to the sample covariance inverse’s MSFE *within each column*, by rolling window size. The sample covariance inverse’s own ratio is trivially 1.000, shown alongside its absolute MSFE (unscaled units) in brackets. Note that ratios only compare methods within a column, not performance across window sizes. Each method’s own absolute MSFE can be recovered by multiplying its ratio with the sample covariance’s absolute MSFE in that column. No bias correction is applied. Bold marks the lowest (best) ratio in each column.

Table 4.1 compares each method’s MSFE against the sample covariance method, within each rolling window size. This comparison is relative and only holds within a window. We return to an absolute comparison across windows later in this section. Within each window, the ranking is largely stable: Adaptive GL performs best throughout, followed by Bayesian GL and Bayesian factor GL, with the sample covariance inverse performing worst at every R . Between Bayesian GL and Bayesian factor GL the order is not fixed: Bayesian GL performs better at $R = 50$ and $R = 60$, but Bayesian factor GL overtakes it at $R = 70$. The results match a feature of the simulation study: the GL-based methods perform comparatively better to the sample covariance inverse once the window size gets closer to the number of forecasters p . The gap here is larger than in the simulation, as $p = 35$ against $R \in \{50, 60, 70\}$ is a tighter ratio than the simulation’s $T = 100$ against p up to 50. In this case, at $R = 50$, Adaptive GL brings MSFE down to about 14% of the sample covariance inverse. The above result makes a case for sparsity: with p closer to R , the sample covariance inverse becomes a worse estimate, and imposing sparsity on Ω provides a clear benefit.

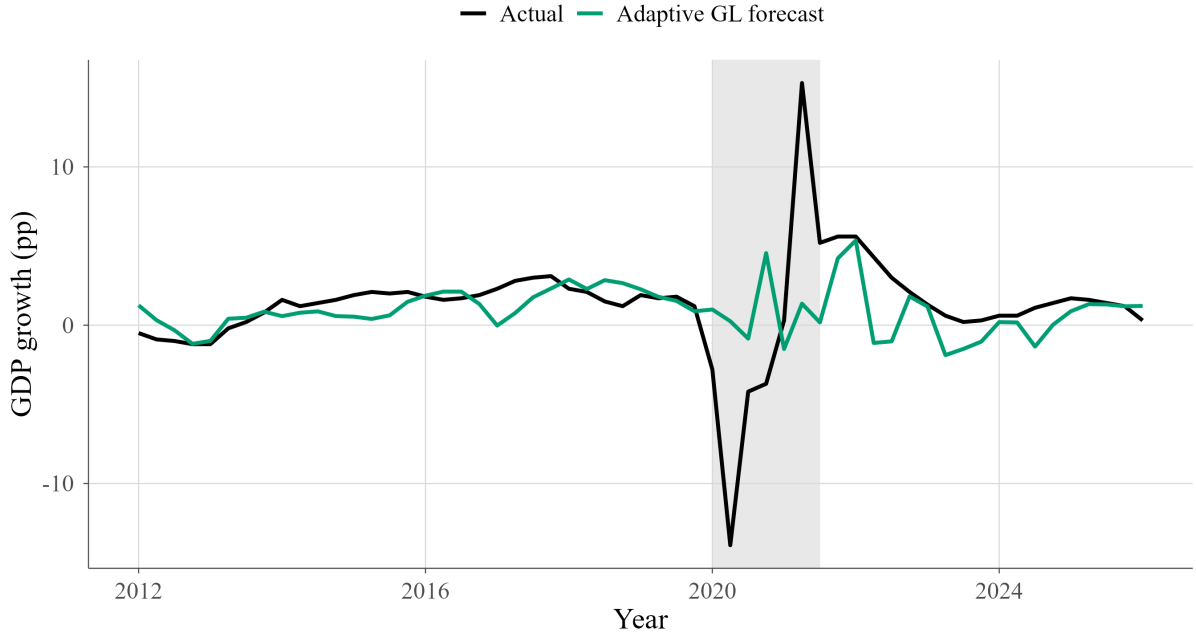


Figure 4.2: Actual GDP growth against the Adaptive GL combined forecast, $R = 50$. The shaded region marks the COVID-19 quarters.

Figure 4.2 puts this ratio in context. Even Adaptive GL, the best-performing method at every window size, tracks actual GDP growth closely until 2020 and then misses both the collapse and the rebound: no linear combination of individual point forecasts could have anticipated a shock that none of the underlying forecasters predicted. More importantly, the break also lands where the method is weakest: with only $R=50$ observations for $p=35$ forecasters, there is little room to deal with a shock this large in estimation, and the forecast combination remains volatile for several years afterwards. The normality assumption adds to this problem: a Gaussian posterior assigns near-zero density to an error this large, so it fails to quantify how uncertain the combination really is during these quarters. Much of every method's MSFE in Table 4.1 is concentrated in these same few COVID-19 quarters.

Compared to the simulation, the Inverted Wishart performs considerably better here relative to the sample covariance inverse. Section 1.2 describes the prior's v pseudo-observations as pulling the posterior toward the scale matrix $\mathbf{S}$. Here $\mathbf{S} = \mathbf{I}_p$, so the prior pulls Σ toward the identity matrix. At $R = 50$, $v = p + 1 = 36$ pseudo-observations are competing against only

$R = 50$ real ones, so the prior carries almost as much weight as the data. In the simulation, $T = 100$ was always well above p , so the same prior mattered far less. A reasonable explanation is that, at R this close to p , the sample covariance itself is so poorly estimated that shrinking it towards the identity still helps, even with a mismatched target. Furthermore, we are now dealing with real data, not the simulation's favourable setup for shrinkage. The gain we therefore see here may not be fully attributable to shrinkage itself, but rather to numerical stability: a sample covariance matrix this close to singular can produce extreme forecasts that dominate the average. The Inverted-Wishart posterior avoids this, since adding the prior's scale matrix keeps it well-conditioned no matter how close to singular the raw data is.

The Bayesian factor GL's relative standing here is close to what our simulation showed at a comparable number of forecasters p . There, Bayesian factor GL's MSE followed a single-peaked shape in p , rising then falling, and by $p \approx 30\text{--}50$ it followed just behind Bayes GL and Adaptive GL. Table 4.1 shows approximately the same ordering at $p = 35$. Since the simulation's DGP has no true factor structure at all, the pattern observed in Table 4.1 cannot be used as evidence that forecasters' errors do or do not share common factors. The same ranking emerges here whether or not that structure is actually present. This matters because whether macroeconomic and time-series forecast errors are better described by common factors or by a non-factor structure remains an open debate in the forecasting literature. The result here is best read as a cost-benefit point: BFGL estimates $\mathbf{\Omega}$ almost as well as Bayes GL and Adaptive GL, but its added complexity from the extra factor-loading step does not translate into any gain over the simpler GL-based methods here. The factor-drifting mechanism identified in Chapter 3 is also still present here, and is discussed later on in this chapter.

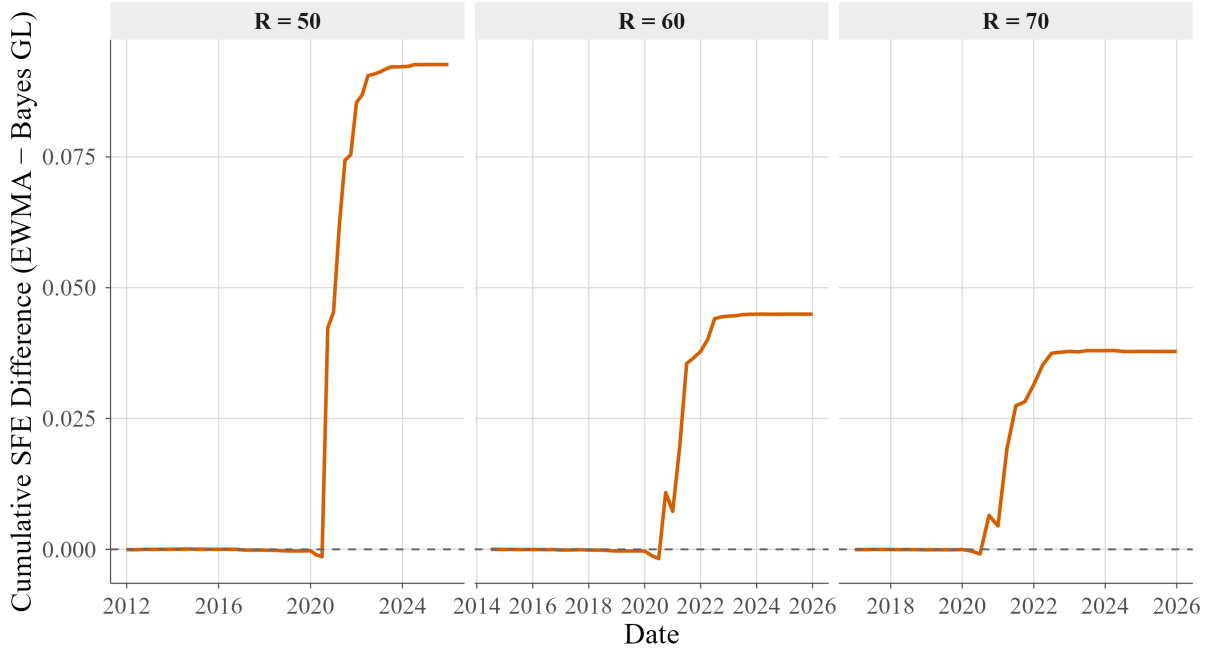


Figure 4.3: Cumulative difference in squared forecast error between the discounted specification (EWMA) and Bayes GL, by rolling window size. As mentioned before, the comparison is isolated to only the standard Bayes GL and its discounted version. At each quarter, we square that quarter’s forecast error for both methods, take the difference (EWMA minus Bayes GL), and add it to a running total. The window then moves one quarter forward and the process repeats. A rising line implies that EWMA has accumulated more error than Bayes GL so far.

EWMA and Bayes GL share the same GL prior and sampler, differing only in whether $\mathbf{S}$ weights all R quarters equally or is replaced by the discounted $\mathbf{S}_t(\delta)$. Figure 4.3 plots their cumulative SFE difference to see where EWMA’s underperformance in Table 4.1 comes from: the line stays flat near zero until around 2020, then rises sharply and flattens again, so almost the entire performance gap arises in this one period. This matches how EWMA is constructed. Discounting only matters when recent quarters differ meaningfully from older ones, and the COVID-19 shock is exactly this case. EWMA’s extra weight on the shock quarters made its estimate of $\mathbf{\Omega}$ more sensitive to them, while Bayes GL’s uniform weighting smooths out the same shock across the full window. This exposes a limitation of EWMA as implemented here: the decay factor δ is fixed for the whole sample and so cannot distinguish a shift in the covariance structure from a temporary shock. A state-space approach such as the dynamic model averaging framework of

Koop and Korobilis [2012], where a Kalman filter updates its state estimate continuously, would likely handle this much better.

A slightly sharper story is that discounting on its own does provide a genuine benefit: it is the COVID-19 shock that complicates the picture. Table 4.2 splits each window’s test periods at 2020Q1 to show this: before 2020, EWMA’s MSFE is slightly below Bayes GL’s at every R , a modest but consistent benefit from discounting. After 2020, the ratio reverses sharply, rising to around 40-70% above Bayes GL.

	$R = 50$	$R = 60$	$R = 70$
EWMA / Bayes GL, pre-2020	0.922	0.909	0.968
EWMA / Bayes GL, post-2020	1.670	1.440	1.387

Table 4.2: MSFE of EWMA relative to Bayes GL within each rolling window size, split at 2020Q1. A ratio below 1 means EWMA outperforms Bayes GL over that sub-period.

The log score in Table 4.3 confirms the MSFE ranking: Adaptive GL leads at every R , followed by Bayes GL, with the sample covariance inverse worst throughout. This agreement is useful because the two metrics test different things: MSFE only evaluates the accuracy of the combined point forecast, whilst the log score evaluates the full predictive density around the point forecast, produced by the Monte Carlo integration over Ω draws. The fact that same sparsity-based methods win on both suggests the benefit of the graphical lasso prior extends beyond point accuracy to the quality of the predictive density.

Method	$R = 50$	$R = 60$	$R = 70$
Sample Cov. Inverse	−39.342	−13.364	−13.451
Inv-Wishart	−3.746	−4.171	−4.894
Bayes GL	−3.325	−3.808	−4.483
Adaptive GL	−3.192	−3.625	−4.050
EWMA	−3.600	−3.749	−4.488
BFGL	−3.818	−3.986	−4.427

Table 4.3: Mean log predictive score by rolling window size. No bias correction is applied. Bold marks the highest (best) score in each column.

As a further check, Appendix A.3 defines and reports PIT histograms for the sample covari-

ance inverse, Adaptive GL, and BFGL. The sample covariance inverse shows a sort of U-shape at every R : realized GDP growth appears in the tails of its predictive density far more often than it should. This means the predictive variance is too narrow, so the method is overconfident. Adaptive GL and BFGL are both closer to uniform, but not perfectly uniform. Incorrect distributional assumptions affect the quality of the predictive density, regardless of how Ω is estimated. In this case, normality (A2) affects all results since it cannot assign a high enough probability to the tails during large shocks like COVID-19. Besides, there is no guarantee that forecast errors are perfectly Gaussian in the first place. Overall, the results match the log score ranking above. Note that with only 37 to 57 test quarters per window, the histograms are based on few observations per bin, so some variability is expected even under good calibration.

Absolute comparison across window sizes

Table 4.1 only compares methods to the sample covariance inverse *within* the same window size. It doesn't tell us about how a method's own performance changes across window sizes. This comparison is in any case not entirely fair across window sizes, since a larger R changes not only how much data each method is estimated on, but also which test quarters end up being forecasted. As depicted in Figure 4.2, this can make a significant difference. A careful look at the absolute MSFE implied by the ratios of Table 4.1 reveals something odd: for almost every method, $R = 60$ outperforms $R = 70$, even though the larger window has more data to work with. Table 4.4 checks whether this is driven by the forecasted test periods or by the window size itself, by restricting all three window sizes to the same 37 test quarters used by $R = 70$. On this common test period, MSFE now falls as R grows for each method. In other words, increasing the window size helps, which is exactly as expected. This confirms that the earlier pattern arose from the test period, not from the window size. When $R = 70$, the test set starts later and has fewer quarters, so the COVID-19 quarters, which are hard to forecast for every method, consists of a bigger share in it. This increases $R = 70$'s average MSFE compared to $R = 60$, even though $R = 70$ uses more data to estimate Ω . Adaptive GL is the one exception with a small non-monotonic result: it still performs best in absolute terms at $R = 60$ rather than at $R = 70$.

Method	$R = 50$	$R = 60$	$R = 70$
Sample Cov. Inverse	0.0186	0.0048	0.0043
Inv-Wishart	0.0044	0.0040	0.0034
Bayes GL	0.0039	0.0029	0.0028
Adaptive GL	0.0025	0.0020	0.0024
EWMA	0.0064	0.0042	0.0038
BFGL	0.0041	0.0030	0.0026

Table 4.4: Absolute MSFE (unscaled units) for $R = 50$, $R = 60$, and $R = 70$, all evaluated on the same 37 test quarters used by $R = 70$. No bias correction is applied.

Bias correction and the rolling window

Section 2.3 relaxes A1, the assumption that forecasters are unbiased. Here, we check how well that correction works in practice, and whether it depends on R . Before we turn to any results, we present some summary statistics of the raw forecast errors in Appendix A.4. The pooled mean error across forecasters and quarters is -0.175 , and 31 out of 35 forecasters show a negative mean error over the full sample. Clearly, assumption A1 does not hold exactly in this panel.

The bias correction of Section 2.3 uses $\hat{\theta}_t$, the sample mean of forecast errors over the same R -quarter window used to estimate Ω . Since $\hat{\theta}_t$ and Ω are estimated from the same window, R affects both at once. Table 4.5 reports each method's own percentage change in MSFE from bias correction, comparing each method only to itself, bias-corrected against raw.

Method	$R = 50$	$R = 60$	$R = 70$
Sample Cov.	-1.95	8.99	9.03
Inv-Wishart	0.29	0.43	0.55
Bayes GL	0.95	0.90	1.03
Adaptive GL	1.37	0.29	0.40
EWMA	1.21	1.58	1.57
BFGL	0.25	0.70	0.65

Table 4.5: Percentage reduction in MSFE from bias correction, by method and R . Positive values mean bias correction lowers MSFE.

Bias correction provides a very modest, stable improvement for the regularised and Bayesian methods, and a larger one for the sample covariance inverse at $R = 60$ and $R = 70$. A possible

reason is that the sample covariance matrix always demeans internally, so that once the forecasts themselves are corrected, its precision estimate and its inputs are consistently bias-free together. The GL-based methods do not estimate $\mathbf{\Omega}$ in a way that is robust to bias, so some residual bias stays in $\mathbf{\Omega}$ even after the forecasts are corrected. This is exactly the kind of gap a Bayesian precision estimator robust to forecaster bias, as suggested in footnote 4, would address in future work.

Three modelling choices

Three practical choices behind the results above are not fixed by the theory in Chapter 2: the EWMA decay rate δ , the factor cap $k_{\max}$ in Bayesian factor GL, and the burn-in length for each MCMC sampler. All three were set beforehand, using economic reasoning or the simulation study rather than the empirical results themselves, to avoid data snooping. The rest of this section explains how each decision was made, then checks whether that reasoning actually holds up against the empirical data.

Sensitivity Analysis: EWMA discount factor

The EWMA method attaches a higher weight to recent quarters in the estimation window through the decay parameter δ . In the above empirical analysis, we have fixed $\delta = 0.97$, which corresponds to a half-life of about 5.7 years. The half-life is defined as the number of years it takes for an observation's weight to fall to half its original value. The choice for δ was made before seeing any results, on the idea that a forecaster's error covariance should track the business cycle: a half-life of roughly one cycle is long enough to smooth out quarterly noise, but short enough to adapt once economic conditions actually change.

To check how this holds up, we reran the empirical application at $R = 60$ across half-lives from two to eight years (Figure 4.4). Both MSFE and log score improve sharply up to about four years, and then quickly flatten out. Our choice of 5.7 years sits well within the flat area. The choice was fixed in advance to avoid data snooping, but looking at it afterwards, 5.7 years turns out to be a sensible choice regardless.

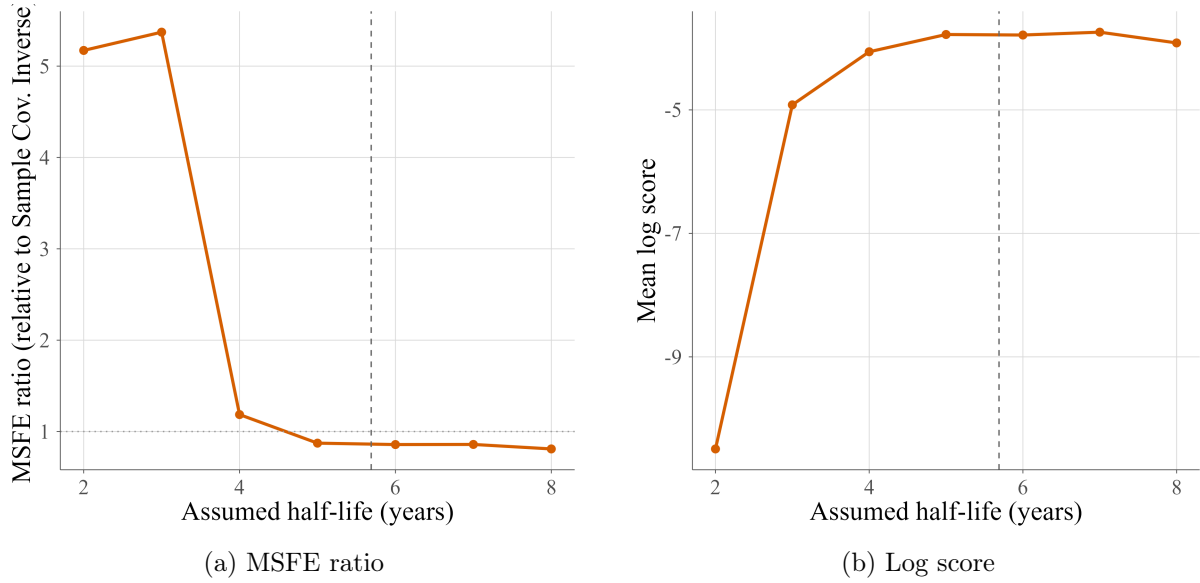


Figure 4.4: Sensitivity of $R = 60$ forecast performance to the assumed EWMA half-life. The dashed line marks the chosen half-life of 5.7 years ($\delta = 0.97$).

Sensitivity Analysis: Factor cap in Bayesian factor GL

The Bayesian factor GL sampler lets the number of latent factors k grow or shrink adaptively, through the birth-death mechanism of the multiplicative gamma process prior. This adaptation happens separately in each rolling window: k is not fixed once for the whole sample, but re-estimated every time the window moves forward one quarter. The empirical application caps this adaptive process at $k_{\max} = 10$, unlike the simulation study which leaves it uncapped. The reasoning is economic: with only $p = 35$ forecasters drawing on broadly the same information set, it is unlikely that their errors share more than a handful of common shocks, let alone twenty or more. The sampler is initialized well below every cap considered here, at $k_{\text{init}} = 5$.

Capping an adaptive sampler like this raises an odd possibility: rather than settling below the cap at whatever number of factors the data actually support, the sampler could instead get pinned exactly at the cap, no matter where it is set. During the simulation study, we noticed k drifting upward in the background, with no cap in place to stop it. [Murphy et al. \[2020\]](#) report very similar behaviour using the same adaptive truncation mechanism (the birth-death rule of [Bhattacharya and Dunson \[2011\]](#)), applied there to small clusters within a mixture model: with

too little data to learn the shrinkage parameters, the number of active factors gets pinned at whatever ceiling is imposed. Their model differs from ours in several ways, but the truncation mechanism itself is identical, and we face the same tight ratio of observations to variables, so this is worth checking here.

We reran the sampler across all 47 rolling windows at $R = 60$ with $k_{\max} \in \{10, 15, 20, 30\}$ (Table 4.6). At $k_{\max} = 10$ and $k_{\max} = 15$, the chain touches the ceiling in every window, and at $k_{\max} = 20$ in nearly all of them. Only at $k_{\max} = 30$ does it stop touching the ceiling in most cases. The cap is therefore doing real work at every level short of 30, not just at $k_{\max} = 10$. Raising it also comes at a cost: mean runtime increases sharply with each step up the grid, while forecasting performance does not improve. If anything, $k_{\max} = 10$ gives the best MSFE ratio of the four.

$k_{\max}$	10	15	20	30
MSFE ratio	0.690	0.735	0.744	0.703
Windows touching cap	100%	100%	93.6%	29.8%
Median k	10	15	20	21
Mean runtime (s)	28.6	56.5	89.7	128.3

Table 4.6: Sensitivity of Bayesian factor GL to the factor cap $k_{\max}$, across all 47 rolling windows at $R = 60$, with the sampler initialised at $k_{\text{init}} = 5$. The MSFE ratio is defined relative to the sample covariance inverse. *Windows touching cap* is the share of windows in which k reaches $k_{\max}$ at least once among the retained post-burn-in draws. *Median k* is the median, across windows, of each window’s own post-burn-in median k .

These results thus imply that $k_{\max} = 10$ is a hard cap. This is a real limitation: with R close to p , there is not enough data per window for the sampler to pick k on its own, and fixing this remains an open problem which we leave for future work. Still, the joint sampler retains a unique advantage: updating the factor loadings and residual precision together gives a coherent posterior, unlike a sequential two-step approach, whether or not k is capped.

Burn-in length for the MCMC methods

Chapter 3 set the simulation burn-in heuristically, then confirmed it with a small trace plot check: 500 iterations for Bayes GL and BFGL, 1000 for Adaptive GL. For the empirical application, the

burn-in for Bayes GL, EWMA, and BFGL was raised to 1500 iterations as a precaution. The real forecast-error panel is harder to estimate from than the simulation DGP, since R is closer to p and the sample includes the 2020 GDP shock. Adaptive GL was left at a 1000 burn-in iterations.

This precaution was checked on the data before analyzing any empirical results. We checked trace plots of Ω_{ij} entries on three $R = 60$ windows: early, late, and one overlapping the COVID-19 shock. All four methods settle well before their burn-in point in every window (Figure 4.5 shows Adaptive GL; the other three are in Appendix A.5 and show a converged pattern, as well as proper mixing after burn-in).

Adaptive GL is the most interesting case. Despite converging cleanly, the trace of the off-diagonal element stays visibly noisier than the other three methods after burn-in, as seen in Figure 4.5. Yet Adaptive GL is also the best-performing method in Table 4.1. Convergence and mixing are answering different questions here: convergence asks whether the chain has forgotten its starting point, mixing asks how quickly it explores the posterior once there. A chain can converge quickly and still mix slowly. Adaptive GL gives each entry its own shrinkage parameter, and the two are updated back and forth: a small entry pushes its shrinkage up, and stronger shrinkage pushes the entry down further. This can cause the chain to get stuck in a heavily-shrunk state for a while before random variation pulls it back out, and the same happens in reverse for a weakly shrunk state. This creates a sort of multimodality, which is what is showing up as noise in the trace plot. It is a side effect of the per-edge shrinkage that makes Adaptive GL effective, not a sign that the sampler has failed to converge.

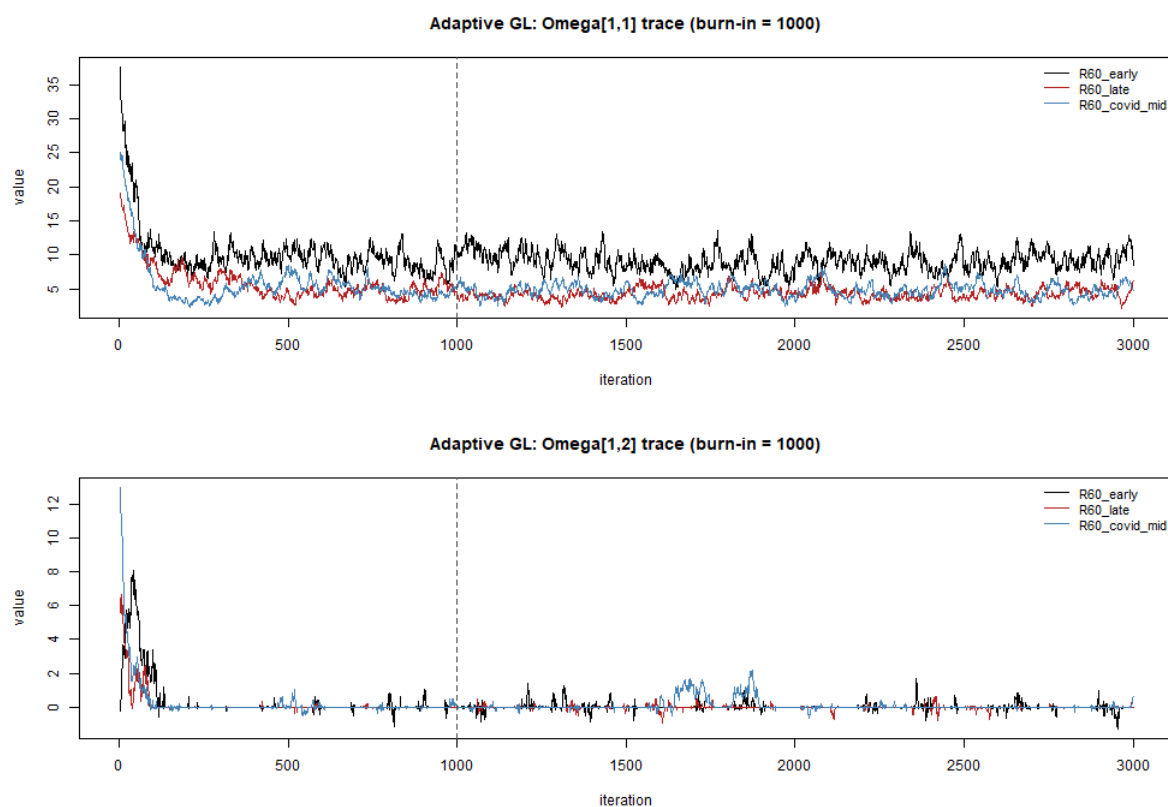


Figure 4.5: Trace plots of Ω_{11} and Ω_{12} for Adaptive GL, across an early, a late, and a COVID-overlap $R = 60$ window.

5 | Conclusion

Forecast combination uses the past accuracy of several individual forecasters or models to build a single, better combined forecast for the future. A Bayesian approach to this problem treats each individual forecast as evidence, updating a prior belief about the unknown value into a posterior predictive distribution. [Winkler \[1981\]](#)'s Bayesian forecast combination framework is unique among existing methods: it resolves reliability, weighting, and uncertainty in a single step, once the precision matrix $\mathbf{\Omega}$ of individual forecast errors is known. This thesis aimed at relaxing some of the assumptions behind that framework and evaluating the consequences, both theoretically and empirically, with a particular focus on high-dimensional forecasting panels.

The assumption that drove most of this thesis' contribution is that $\mathbf{\Omega}$ is known. We did not relax this assumption in the strict sense of treating $\mathbf{\Omega}$ as unknown and integrating it out analytically. Instead, we interpreted "known" pragmatically: Winkler's weights formula only needs some value of $\mathbf{\Omega}$ at the point where weights are computed, so any estimate can play that role. This flexibility let the framework use modern advances in high-dimensional precision estimation, which can help a forecaster combine a large panel of individual forecasts into a single predictive distribution.

The graphical lasso is one such advance: by treating many pairs of forecasters as conditionally independent, it fixes many off-diagonal entries of $\mathbf{\Omega}$ at exactly zero, which reduces the number of free parameters to estimate in $\mathbf{\Omega}$. We used the Bayesian version of the graphical lasso, introduced by [Wang \[2012\]](#), rather than the frequentist counterpart, because it returns a full posterior over $\mathbf{\Omega}$ rather than a single point estimate. This matters exactly in a high-dimensional setting: once the panel of forecasters p approaches the amount of data T available to estimate $\mathbf{\Omega}$, a point estimate is itself highly uncertain, and treating it as fixed risks understating uncertainty in the final combination. Drawing many values of $\mathbf{\Omega}$ from its posterior distribution, and averaging the resulting forecast combinations over them, lets this estimation uncertainty carry through to the final predictive distribution. It is also the contrast to our closest related work: [Lee and Seregina \[2025\]](#) use the same technique to estimate $\mathbf{\Omega}$ in a large panel, but their approach is frequentist and returns only a point estimate.

Both the simulation study and the empirical application pointed to the same conclusion: sparsity on Ω pays off once the number of forecasters p grows closer to the amount of data available to estimate it. In the simulation, the gap between the GL-based methods and the ‘basic’ sample covariance inverse widens as p grows towards $T = 100$. In the empirical application, where we combined Euro GDP growth forecasts from $p = 35$ professional forecasters, the ratio between the number of forecasters and available data is even smaller, with rolling window sizes of $R \in \{50, 60, 70\}$. The advantage of sparsity became correspondingly larger: Adaptive GL, one of the graphical lasso methods and the best-performing one across both applications, brings the MSFE of the final forecast combination down to about 14% of the sample covariance inverse at $R = 50$.

Not every extension provided such a clear result. The introduced Bayesian factor GL adds a factor-extraction step before applying a GL prior, intended to capture common shocks across forecast errors before imposing sparsity on what remains. In the simulation, where no true factor structure exists, this step consistently added noise rather than removing it. In the empirical application, Bayesian factor GL performed almost as well as the standard GL methods, but not better. The added complexity of the joint sampler therefore provides no clear forecasting gains. Given the simulation result, this ranking alone cannot tell us whether forecaster errors are actually driven by common macroeconomic shocks, because the same ranking emerged when they are not.

The remaining two extensions further relaxed assumptions of Winkler’s framework: that forecasters are unbiased, and that the error covariance is stationary over time. Bias correction, which relaxes the first, subtracts an estimated bias from each forecaster’s point forecast before any weighting is applied, but does not change how Ω itself is estimated. It still provided a modest, consistent improvement for every regularized method, and a larger one for the sample covariance inverse, given that the sample covariance demeanes internally. EWMA discounting, which relaxes the second assumption by letting Ω vary over time, provided a clear benefit in ordinary quarters, outperforming its stationary counterpart before 2020 at every window size. Once the COVID-19 shock enters the estimation window, however, it overweighted that shock relative to a uniformly-weighted alternative, which is enough to get rid of all previous bene-

fits. Both methods could benefit further from more nuanced approaches, such as a bias-proof precision estimator or a state-space model to improve the discounting mechanism.

A few limitations follow directly from our choices. Normality is never relaxed, even though the COVID-19 shock and other extreme quarters strongly affect the empirical results. Missing SPF responses are imputed before estimation, so this uncertainty never reaches the posterior distribution of Ω . Computational cost is a concern too: every method introduced except the sample covariance inverse and the Inverted Wishart prior requires posterior sampling, and this cost grows with panel size across all of them. Bayesian factor GL is most affected, since it samples factor loadings and factor scores next to the GL updates. The Bayesian factor GL's truncation mechanism is also a limitation: it drifts towards its ceiling in the empirical application, making $k_{\max} = 10$ a hard constraint rather than a safeguard, and raising the cap only pushes the ceiling further out at high computational cost.

Future work could pursue several of these open points directly. A Bayesian precision matrix estimator robust to forecaster bias, rather than treating bias correction and precision estimation as two separate steps, would close the gap identified in Section 4.2. A state-space extension of the discounting mechanism, closer to the dynamic model averaging framework of [Koop and Korobilis \[2012\]](#), would let the effective window adapt to shifts in the error covariance rather than reacting uniformly to each recent observation. A way of choosing $k_{\max}$ automatically, rather than fixing it by hand, would improve the functioning of the Bayesian factor GL under many forecasters and limited data. Relaxing normality, for instance by replacing the Gaussian likelihood with a heavy-tailed distribution, would let the framework accommodate shocks like COVID-19.

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A | Supplementary Material

A.1 Code Reproduction

All R code used to produce the results in this thesis is available at https://github.com/nlblom/thesis_reproduction.

All experiments were run using R version 4.3.2 on Windows 11, with **BayesianGLasso** version 0.2.0 and **abglasso** version 0.1.1. Runtimes reported in this thesis were measured on an Intel Core i7-13700H CPU (14 cores).

A.2 Derivation of the full conditional for $\boldsymbol{\lambda}$

This section derives Equation (2.22), the joint full conditional distribution of $\boldsymbol{\lambda}$, which is the vector stacking all p rows of $\mathbf{\Lambda}$. In the directed acyclic graph implied by the model, $\boldsymbol{\lambda}$'s full conditional depends only on its Markov blanket: its own prior, and the likelihood of its child $\mathbf{U}$. This yields $p(\boldsymbol{\lambda} \mid \mathbf{F}, \boldsymbol{\Omega}_\varepsilon, \mathbf{U}) \propto p(\mathbf{U} \mid \boldsymbol{\lambda}, \mathbf{F}, \boldsymbol{\Omega}_\varepsilon) \cdot p(\boldsymbol{\lambda})$, with $p(\mathbf{U} \mid \boldsymbol{\lambda}, \mathbf{F}, \boldsymbol{\Omega}_\varepsilon)$ the likelihood and $p(\boldsymbol{\lambda})$ the MGP prior. We start from the likelihood, isolate the terms depending on $\boldsymbol{\lambda}$, and then combine these with the prior.

The model assumes $\boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, \boldsymbol{\Omega}_\varepsilon^{-1})$ independently across $t = 1, \dots, T$. This is the only distributional assumption the model makes directly, so we begin here and later re-express the result in terms of $\mathbf{U}$ and $\mathbf{\Lambda}$, which will provide us the likelihood that is needed above. Using the standard multivariate Gaussian log-density, and dropping the normalizing constant that does not depend on $\boldsymbol{\varepsilon}_t$,

$$\log p(\mathbf{E} \mid \boldsymbol{\Omega}_\varepsilon) \propto -\frac{1}{2} \sum_{t=1}^T \boldsymbol{\varepsilon}_t' \boldsymbol{\Omega}_\varepsilon \boldsymbol{\varepsilon}_t,$$

where $\mathbf{E}$ is the $T \times p$ matrix with rows $\boldsymbol{\varepsilon}_t'$. Since $\boldsymbol{\varepsilon}_t' \boldsymbol{\Omega}_\varepsilon \boldsymbol{\varepsilon}_t$ is a scalar, it does not change under the trace operator, and the cyclic property of the trace¹ gives:

¹For matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ of compatible dimensions for matrix multiplication, $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$.

$$\sum_{t=1}^T \varepsilon_t' \Omega_\varepsilon \varepsilon_t = \sum_{t=1}^T \text{tr}(\varepsilon_t' \Omega_\varepsilon \varepsilon_t) = \sum_{t=1}^T \text{tr}(\Omega_\varepsilon \varepsilon_t \varepsilon_t') = \text{tr} \left(\Omega_\varepsilon \sum_{t=1}^T \varepsilon_t \varepsilon_t' \right) = \text{tr}(\Omega_\varepsilon \mathbf{E}' \mathbf{E}),$$

using $\sum_t \varepsilon_t \varepsilon_t' = \mathbf{E}' \mathbf{E}$. Substituting $\mathbf{E} = \mathbf{U} - \mathbf{F} \boldsymbol{\Lambda}'$ (valid without a Jacobian correction, since this is just a linear shift) yields:

$$\log p(\mathbf{U} \mid \boldsymbol{\Lambda}, \mathbf{F}, \Omega_\varepsilon) \propto -\frac{1}{2} \text{tr} [\Omega_\varepsilon (\mathbf{U} - \mathbf{F} \boldsymbol{\Lambda}')' (\mathbf{U} - \mathbf{F} \boldsymbol{\Lambda}')]. \quad (\text{A.1})$$

Or equivalently, using that $\text{tr}(\Omega_\varepsilon \mathbf{E}' \mathbf{E}) = \text{vec}(\mathbf{E})' (\Omega_\varepsilon \otimes \mathbf{I}_T) \text{vec}(\mathbf{E})$, the log-likelihood is proportional to:

$$\log p(\mathbf{U} \mid \boldsymbol{\Lambda}, \mathbf{F}, \Omega_\varepsilon) \propto -\frac{1}{2} \text{vec}(\mathbf{E})' (\Omega_\varepsilon \otimes \mathbf{I}_T) \text{vec}(\mathbf{E}).$$

Let $\boldsymbol{\lambda} = \text{vec}(\boldsymbol{\Lambda}')$ be the vector stacking the p rows $\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_p$ of $\boldsymbol{\Lambda}$. Writing $\mathbf{U} = \mathbf{F} \boldsymbol{\Lambda}' + \mathbf{E}$ as a multivariate regression and vectorizing with $\text{vec}(\mathbf{F} \boldsymbol{\Lambda}') = (\mathbf{I}_p \otimes \mathbf{F}) \boldsymbol{\lambda}$ yields $\text{vec}(\mathbf{E}) = \text{vec}(\mathbf{U}) - (\mathbf{I}_p \otimes \mathbf{F}) \boldsymbol{\lambda}$ so that

$$\text{vec}(\mathbf{E})' (\Omega_\varepsilon \otimes \mathbf{I}_T) \text{vec}(\mathbf{E}) = [\text{vec}(\mathbf{U}) - (\mathbf{I}_p \otimes \mathbf{F}) \boldsymbol{\lambda}]' (\Omega_\varepsilon \otimes \mathbf{I}_T) [\text{vec}(\mathbf{U}) - (\mathbf{I}_p \otimes \mathbf{F}) \boldsymbol{\lambda}].$$

Expanding this quadratic form in $\boldsymbol{\lambda}$, the term not involving $\boldsymbol{\lambda}$ drops out as a constant, leaving a quadratic term $\boldsymbol{\lambda}' (\mathbf{I}_p \otimes \mathbf{F})' (\Omega_\varepsilon \otimes \mathbf{I}_T) (\mathbf{I}_p \otimes \mathbf{F}) \boldsymbol{\lambda}$ and a linear term $-2 \boldsymbol{\lambda}' (\mathbf{I}_p \otimes \mathbf{F})' (\Omega_\varepsilon \otimes \mathbf{I}_T) \text{vec}(\mathbf{U})$. Applying the mixed-product property $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD})$ twice, the matrix scaling the quadratic term becomes:

$$(\mathbf{I}_p \otimes \mathbf{F})' (\Omega_\varepsilon \otimes \mathbf{I}_T) (\mathbf{I}_p \otimes \mathbf{F}) = \Omega_\varepsilon \otimes (\mathbf{F}' \mathbf{F}),$$

and applying the mixed-product property once more, followed by the identity $(\mathbf{A} \otimes \mathbf{B}) \text{vec}(\mathbf{X}) = \text{vec}(\mathbf{B} \mathbf{X} \mathbf{A}')$, together with $\Omega_\varepsilon' = \Omega_\varepsilon$ by symmetry, the linear term's coefficient becomes

$$(\mathbf{I}_p \otimes \mathbf{F})'(\boldsymbol{\Omega}_\varepsilon \otimes \mathbf{I}_T)\text{vec}(\mathbf{U}) = (\boldsymbol{\Omega}_\varepsilon \otimes \mathbf{F}')\text{vec}(\mathbf{U}) = \text{vec}(\mathbf{F}'\mathbf{U}\boldsymbol{\Omega}_\varepsilon).$$

Combining these two results, the log-likelihood in terms of $\boldsymbol{\lambda}$ is

$$\log p(\mathbf{U} \mid \boldsymbol{\lambda}, \mathbf{F}, \boldsymbol{\Omega}_\varepsilon) = \text{const} - \frac{1}{2} \boldsymbol{\lambda}' [\boldsymbol{\Omega}_\varepsilon \otimes (\mathbf{F}'\mathbf{F})] \boldsymbol{\lambda} + \boldsymbol{\lambda}' \text{vec}(\mathbf{F}'\mathbf{U}\boldsymbol{\Omega}_\varepsilon).$$

Since each λ_{jh} is independently $N(0, \phi_{jh}^{-1} \tau_h^{-1})$ distributed under the MGP prior, stacking the precisions to create a prior yields $\boldsymbol{\lambda} \sim N(\mathbf{0}, \mathbf{D}^{-1})$, where $\mathbf{D} = \text{blockdiag}(\mathbf{D}_1^{-1}, \dots, \mathbf{D}_p^{-1})$ and $\mathbf{D}_j^{-1} = \text{diag}(\phi_{j1}\tau_1, \dots, \phi_{jq}\tau_q)$. This gives $\log p(\boldsymbol{\lambda}) = \text{const} - \frac{1}{2} \boldsymbol{\lambda}' \mathbf{D} \boldsymbol{\lambda}$. Adding this to the log-likelihood above gives an expression of the form $\text{const} - \frac{1}{2} \boldsymbol{\lambda}' \mathbf{Q} \boldsymbol{\lambda} + \boldsymbol{\lambda}' \mathbf{b}$ with $\mathbf{Q} = \mathbf{D} + \boldsymbol{\Omega}_\varepsilon \otimes (\mathbf{F}'\mathbf{F})$ and $\mathbf{b} = \text{vec}(\mathbf{F}'\mathbf{U}\boldsymbol{\Omega}_\varepsilon)$. Adding the prior and likelihood is a standard Gaussian-conjugacy result: a Gaussian prior combined with a quadratic log-likelihood always gives a Gaussian posterior, obtained by completing the square. This yields the conjugate posterior of Equation (2.22):

$$\boldsymbol{\lambda} \mid \mathbf{F}, \boldsymbol{\Omega}_\varepsilon, \mathbf{U} \sim N_{pq}(\mathbf{Q}^{-1}\mathbf{b}, \mathbf{Q}^{-1}), \quad \mathbf{Q} = \mathbf{D} + \boldsymbol{\Omega}_\varepsilon \otimes (\mathbf{F}'\mathbf{F}), \quad \mathbf{b} = \text{vec}(\mathbf{F}'\mathbf{U}\boldsymbol{\Omega}_\varepsilon).$$

A.3 PIT Calibration

The probability integral transform (PIT) is the CDF of the predictive density evaluated at the realized outcome, $\text{PIT}_t = F_t(y_t)$, where F_t is the predictive CDF implied by the Monte Carlo mixture at period t . If the predictive densities are well calibrated, the PIT values should be uniformly distributed on $[0, 1]$.

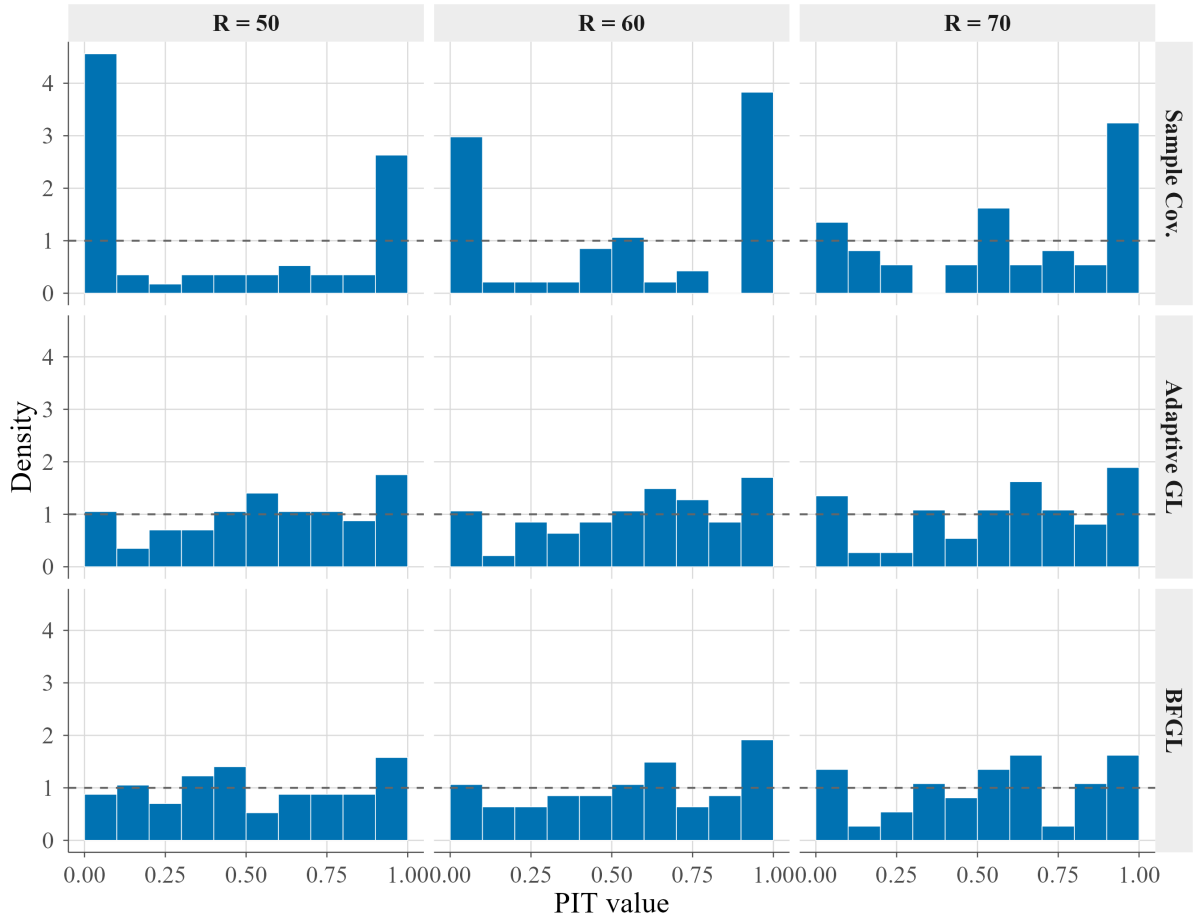


Figure A.1: Probability integral transform (PIT) histograms by rolling window size R , for the sample covariance inverse, Adaptive GL, and BFGL. We limit this plot to the naive baseline, the best-performing method, and the introduced factor-method.

A.4 Summary Statistics of Forecast Errors

Table A.1 reports summary statistics of the raw forecast errors $u_{it} = y_t - f_{it}$, in percentage points, before any bias correction is applied. Most forecasters are negatively biased: 31 of the 35 forecasters have a negative mean error over the full sample, meaning that their point forecasts were on average too high relative to realized GDP growth.

Statistic	Value (pp)
Pooled mean error	-0.175
Pooled SD of error	2.236
Median forecaster mean error	-0.178
Min forecaster mean error	-0.414
Max forecaster mean error	0.184

Table A.1: Summary statistics of forecast errors $u_{it} = y_t - f_{it}$, pooled across all $p = 35$ forecasters and $T = 107$ quarters. The last three rows describe the distribution, across forecasters, of each forecaster's own mean error over the full sample.

A.5 Additional Burn-in Trace Plots

Figures A.2–A.4 show the same diagnostic as Figure 4.5 in Chapter 4, for Bayes GL, EWMA, and BFGL. All three settle cleanly before their burn-in point in every window, and provide good mixing afterwards.

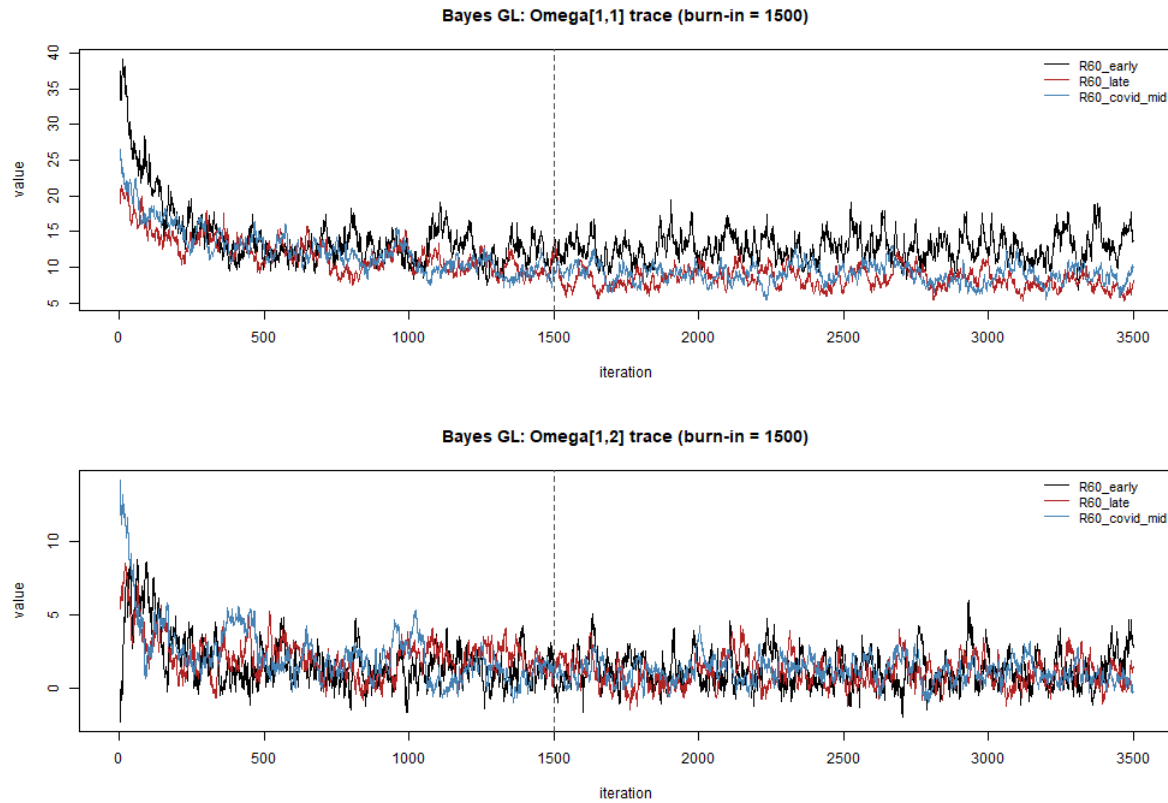


Figure A.2: Trace plots of Ω_{11} and Ω_{12} for Bayes GL, across an early, a late, and a COVID-overlap $R = 60$ window.

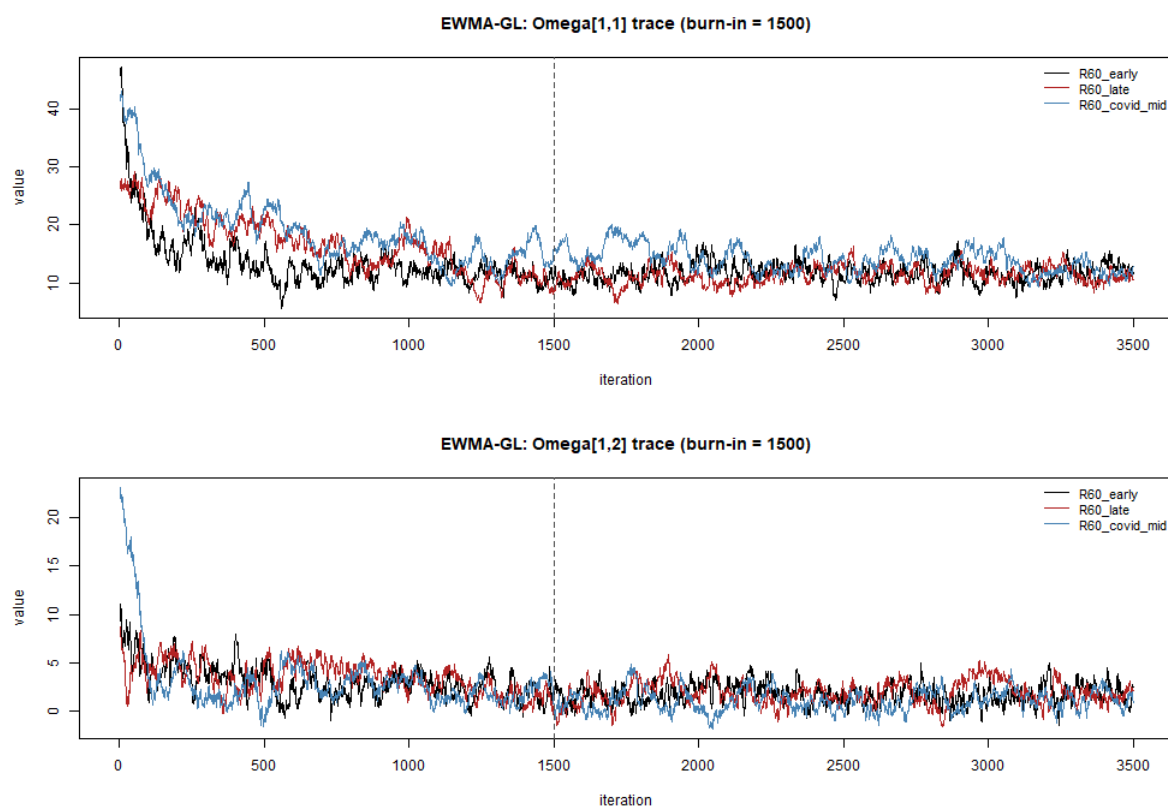


Figure A.3: Trace plots of Ω_{11} and Ω_{12} for EWMA-GL, across an early, a late, and a COVID-overlap $R = 60$ window.

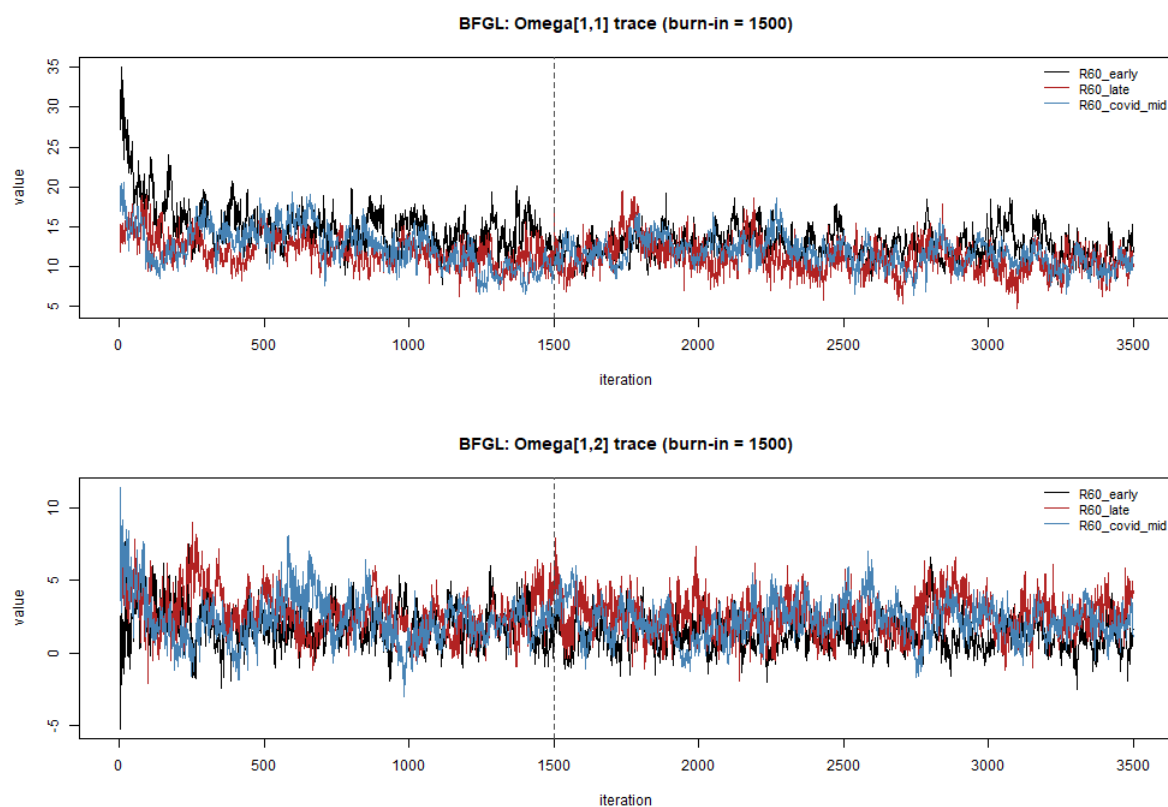


Figure A.4: Trace plots of Ω_{11} and Ω_{12} for BFGL, across an early, a late, and a COVID-overlap $R = 60$ window.

